

Synopsis: A Scripting Language for Sufficient Alignment in Genomics

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Abstract

Genomic comparison is conducted almost entirely in the vocabulary of *exact* correspondence: percent identity, edit distance, alignment score. This vocabulary is not merely imprecise for the question practitioners actually ask — *do these two things do the same thing?* — it is provably unable to answer it. We prove two separation results. There exist pairs of units whose content is separated at the theoretical minimum, so that no content-based criterion can distinguish them, yet whose effect on the surrounding network diverges without bound; and there exist pairs whose content is arbitrarily expensive to reconcile yet whose effect is identical. Content comparison collapses the first class to *same* and the second to *different*, and is wrong in both directions. The two classes are orthogonal axes of a plane that a scalar criterion projects onto a line.

We construct the theory needed to occupy that plane. Working in finite weighted graphs with strictly positive weights, we establish a *contact floor*: a strictly positive lower bound β on the cost of any distinction. We show that identity is borne by a minimum cut — a region of edges — and never by a vertex or a label, so that it survives arbitrary relabelling exactly and is not carried by any name a database can store. We show that the value a unit registers is a function of the pair (unit, receiver) and not of the unit, and that the resulting disagreement between receivers is not measurement error. We then give a bidirectional comparison, *cross-*

demand, that certifies correspondence without reading meaning, and prove that its relaxation terminates in at most $\lceil D_0/\eta \rceil$ steps or declines — there is no third outcome and no divergence.

From this theory we design and specify **synopsis** (extension `.syp`), a total scripting language for genomic experiments. Its name is taken from the synoptic gospels, the books that are read in parallel columns because they *align sufficiently*, not because they agree word for word; the discipline of reading them is exactly the discipline we formalise. Synopsis is designed around refusals. Sequences are opaque and cannot be indexed. Every value carries a residue bounded below by the floor, and no constructor produces zero residue, so **identical** is not expressible. The verdict constructor is *4-ary*: omitting the response columns is an arity error, not a weaker result. Comparison across receiver frames is a scope error. Thresholds have no defaults. Iteration is bounded by construction and relaxation typechecks only when its step floor is bounded below, in which case the compiler emits the termination bound as a proof obligation it has discharged. We prove strong normalisation, floor preservation, scope safety, and four *refusal theorems* stating that certain scientifically invalid claims have no well-typed representation.

We give the full grammar, a typing judgement, a small-step operational semantics, and worked programs reproducing three families of genomic experiment — matched-filter motif detection, spectral homology search over a database, and four-column functional adjudi-

cation — in a single language whose output is a report bearing certificates rather than a number. We report honestly that one assumption the four-column verdict requires, the independence of the verdict from the choice of admissible response, *fails* on solved biochemical networks: only 28.6% of comparisons fall within threshold, with a maximum discrepancy of 0.307 against $\theta = 0.01$. Synopsis makes this failure syntactically visible: the response operator requires its map to be named, so no program can conceal which arbitrary choice it made. We regard designing a language that cannot hide its open problems as the point.

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1 Introduction

1.1 Two failures of the same kind

Consider two questions a working geneticist asks daily.

- (i) **A synonymous variant.** A single base changes; the codon changes; the amino acid does not. Every content-based comparison reports the protein product as identical. Yet synonymous variants demonstrably alter translation kinetics, co-translational folding, mRNA stability and substrate specificity [46, 50, 33]. The comparison that reports *same* is not approximately right. It is reporting on a quantity that was never the one at issue.
- (ii) **Two unrelated enzymes catalysing one reaction.** Non-homologous isofunctional enzymes are common enough to have a name and a systematic census [41, 20]. Their sequences share nothing; every alignment reports maximal divergence. They do the same job. The comparison that reports *different* is, again, answering a different question than the one asked.

These are usually filed as separate difficulties — one a false positive, one a false negative — to be patched by better substitution matrices, deeper profile models, or structural prediction [24, 15, 29]. We will show they are one difficulty, and that it is structural rather than statistical.

The claim we prove is sharp. Let *content divergence* be any quantity computed from the two units in isolation, and let *response divergence* be a quantity computed from what each unit does to the network it sits in. Theorem 5.1 constructs a family in which content divergence is pinned at exactly zero — not small, exactly zero, at the theoretical minimum — while response divergence grows without bound. Theorem 5.4 constructs a family in which response divergence is exactly zero while content divergence grows. The two families lie on orthogonal axes of a two-dimensional plane. Any criterion returning a scalar computed from content alone projects that plane onto one axis, and necessarily maps one entire family to the origin. No

refinement of the scalar repairs this; the defect is one of dimension, not of accuracy.

1.2 Sufficient alignment

The response to a comparison that cannot be exact is not to abandon comparison. It is to change what one asks of it.

There is an old and instructive precedent. The first three canonical gospels are called *synoptic* — seen together — because they can be laid out in parallel columns and read across [23, 58]. They are not identical. They differ in wording, in order, in detail, and in what each chooses to omit. A concordance demanding word-for-word agreement finds them in constant contradiction and produces nothing. The synoptic reading demands something weaker and far more useful: that the columns correspond *sufficiently* for a passage in one to be located against a passage in another, with the residue — what does not correspond — retained and displayed rather than discarded. The discipline is not to eliminate the difference. It is to bound it, to know where it lies, and to be able to say when it is small enough not to matter for the question at hand.

That is precisely the discipline genomic comparison needs and does not have. We take the name of our language from it. A *synopsis* is a parallel-column reading in which correspondence is certified, residue is carried, and the verdict is about sufficiency rather than identity.

The choice of name commits us to three design consequences, which we discharge in Section 6 and prove in Section 9.

- (a) **Residue is never discarded.** A comparison that returns a single number has thrown the residue away. In synopsis every value carries its residue in its type, and there is no constructor producing zero residue (Theorem 9.4).
- (b) **Exactness is not expressible.** If *identical* can be written, it will be written, and the two failures of Section 1 return. In synopsis it has no constructor (Theorem 9.5).
- (c) **The columns are structural.** A synopsis with one column is not a synopsis.

The verdict constructor of synopsis takes four columns and omitting a pair is an arity error, not a permitted simplification (Theorem 9.6).

1.3 Where the invariant is not

Before building, we should say plainly what we take to be the source of the trouble, because it determines everything about the language’s treatment of sequence.

The tacit premise of content comparison is that whatever makes two units functionally the same is *written in them*, so that a sufficiently clever reading of the two strings recovers it. We claim this premise is false, and we can make the claim precise rather than rhetorical.

Remark 1.1 (The invariant is not borne by the string). Theorem 2.7 shows that the quantity preserved under relabelling of a contact structure is the weight of a minimum cut, and Theorem 3.1 shows that this cut is in general a region of edges containing several vertices, not a single vertex. Theorem 2.8 then shows that vertex labels may be permuted arbitrarily without changing any separation cost. A sequence is a labelling. It is therefore, by Theorem 2.8, precisely the part of the structure that the invariant is indifferent to. Reading harder does not help because there is nothing further in the reading.

There is an elementary observation in the same direction that requires no theory at all. Every double-stranded nucleic acid carries two strands [63]. Read the complement in its own $5' \rightarrow 3'$ direction and one obtains a different string, with a different codon decomposition, and different open reading frames. Both strands are physically present; both are read by the cell in different contexts; neither is privileged by the chemistry. If the functional invariant were a property of the string, the molecule would carry two of them and they would disagree. What the molecule actually determines is the *pairing* — the contact structure — and both strings are labellings of it. The invariant, if there is one, lives in the contacts.

Accordingly, in synopsis a sequence is an **opaque** type. It has no indexing operator, no substring operator, no base accessor, and no equality. It enters computation only by projec-

tion into a structure that is a legitimate bearer of the invariant. This is the single most restrictive decision in the language and Section 6.1 defends it in detail.

1.4 Why a language, and why a total one

One could implement all of this as a library. We give three reasons not to.

First, a library cannot refuse. The results of Section 5 say that certain comparisons are invalid — comparing content alone, comparing across incompatible receiver frames, reporting exact identity. In a library these are discouraged by documentation. In a language they can be made unrepresentable. That is the difference between a convention and a theorem, and Section 9 proves four theorems of exactly this shape. A refusal that the typechecker enforces is worth more than a paragraph in a manual that a hurried user does not read.

Second, defaults are where science goes wrong. A library function with a default threshold is an experiment with an undeclared parameter. Multiplied across a pipeline this is the mechanism by which analytic flexibility silently manufactures results [54, 21]. Synopsis has *no default thresholds anywhere*: every numeric parameter that can change a verdict must be written in the program text, and the report prints it. This costs the user keystrokes and buys the reader the ability to see what was chosen.

Third, an experiment should terminate. A laboratory protocol that might not finish is not a protocol. Synopsis is *total*: every well-typed program halts, and we prove it (Theorem 9.1). This is purchased by giving up general recursion and unbounded loops [60], which turns out to cost nothing — none of the genomic procedures we survey needs them — and by requiring that the one genuinely iterative construct, relaxation, be given a step floor, at which point the compiler emits the iteration bound as a discharged obligation (Section 7.6).

This last point drives the one design decision we should flag explicitly because a reader may expect otherwise. Synopsis is an **analytic** language, not a generative one. It evaluates and certifies claims about given material. It

does *not* search a space for sequences meeting a target specification. Generative search requires a solver whose termination depends on the geometry of the search space and cannot be established by typing; admitting it would forfeit Theorem 9.1, which is the strongest property the language has. We judge the trade correct: a total language that certifies is more useful to an experimentalist than a partial one that also proposes, and the generative layer, if wanted, can be built above synopsis and made to emit synopsis programs for certification.

1.5 Contributions and organisation

- Section 2 develops contact structures and proves the contact floor (Theorem 2.4), the invariance of cut weight (Theorem 2.7), and its conditional survival under expansion (Theorem 2.13), with an explicit counterexample (Theorem 2.10) showing the unconditional statement to be *false*.
- Section 3 proves that identity is regional (Theorem 3.1), that labels are free (Theorem 2.8), and that registered value is receiver-relative (Theorem 3.5) with disagreement that is not noise (Theorem 3.6).
- Section 4 builds cross-demand and relaxation and proves monotonicity (Theorem 4.9), the termination dichotomy (Theorem 4.10) and the form of quiescence (Theorem 4.11).
- Section 5 proves the two separation results (Theorem 5.1, Theorem 5.4) and derives the four-column criterion (Theorem 5.10).
- Section 6 states the eleven design rules of synopsis and argues each from a result above.
- Section 7 gives the concrete language: lexis, grammar, types and the typing judgement.
- Section 8 gives the operational semantics.
- Section 9 proves the metatheory: totality, floor preservation, scope safety, and the four refusal theorems.
- Section 10 gives worked `.syp` programs for three families of experiment.
- Section 11 specifies the report and its certificates.
- Section 13 reports the negative result on response independence and what the language does about it.

2 Contact structures and the floor

We build the theory from nothing. A reader who wants only the language may read Section 6 onwards and refer back, but the design rules are consequences of these results and are arbitrary without them.

2.1 Finite weighted graphs

Definition 2.1 (Finite weighted graph). A *finite weighted graph* is a triple $\mathcal{G} = (V, E, w)$ with V a finite non-empty set, $E \subseteq \binom{V}{2}$, and $w : E \rightarrow \mathbb{R}_{\geq 0}$. Weights are *strictly* positive: an edge of weight zero is not a light edge, it is the absence of an edge, and we insist on the distinction because every result below turns on it.

Definition 2.2 (Cut). For $S \subseteq V$ the *cut* induced by S is $\partial S = \{e \in E : |e \cap S| = 1\}$ with weight $\text{cut}(S) = \sum_{e \in \partial S} w(e)$. For $u \neq v$ a set S is *(u, v)-separating* if $u \in S$ and $v \notin S$, and

$$\sigma(u \mid v) = \min\{\text{cut}(S) : S \text{ is } (u, v)\text{-separating}\}.$$

The minimum is over a finite non-empty collection, hence attained.

We call $\sigma(u \mid v)$ the *separation cost*: the cheapest way, in the currency the weights supply, to place u and v on opposite sides of a boundary. By the max-flow min-cut theorem it equals the maximum flow from u to v [17], and is computable in polynomial time [57, 31]; all-pairs values admit a Gomory–Hu tree [22]. We will not need efficiency for the theory, but the language’s cost model does (Section 8.4).

Definition 2.3 (Contact structure). A *contact structure* is a finite weighted graph with a distinguished vertex $\mathbf{m} \in V$, the *medium*, adjacent to every other vertex. Vertices other than $\mathbf{m}$ are *items*, written $I = V \setminus \{\mathbf{m}\}$. We write

$\sigma(v) = \sigma(v \mid \mathbf{m})$ for the *separation cost of an item*, $\Omega = \sum_{e \in E} w(e)$ for the total weight, and

$$\beta(\mathcal{G}) = \min_{e \in E} w(e)$$

for the *contact floor*.

The medium formalises the background against which anything is individuated. Every distinction is a distinction *from* something; $\mathbf{m}$ is that something, and its universal adjacency says the background is always available and never absent. This is the graph-level statement of a very old idea — that a thing is known by what it is not — and it is what makes $\sigma(v)$ the cost of individuating v at all, rather than merely of separating it from a chosen rival.

2.2 The floor

Theorem 2.4 (Contact floor). *Let $\mathcal{G}$ be a contact structure and v an item. Then*

$$\sigma(v) \geq \beta(\mathcal{G}) > 0.$$

Proof. E is finite and non-empty (the medium is adjacent to every item and $I \neq \emptyset$), and w takes values in $\mathbb{R}_{>0}$, so $\beta(\mathcal{G}) = \min_e w(e)$ is the minimum of a finite set of strictly positive reals and is therefore strictly positive.

Let S be any $(v, \mathbf{m})$ -separating set. Since $v \in S$ and $\mathbf{m} \notin S$ and $\{v, \mathbf{m}\} \in E$ by adjacency of the medium, we have $|\{v, \mathbf{m}\} \cap S| = 1$, so $\{v, \mathbf{m}\} \in \partial S$. Hence $\partial S \neq \emptyset$ and $\text{cut}(S) \geq w(\{v, \mathbf{m}\}) \geq \beta(\mathcal{G})$. Taking the minimum over separating S gives $\sigma(v) \geq \beta(\mathcal{G})$. $\square$

The proof is three lines and the content is substantial: *no distinction is free*. There is a strictly positive price on telling any two things apart, fixed by the cheapest contact in the structure. Two items may be as similar as the structure permits — they may sit exactly at the floor — but they cannot be at zero, because zero is not in the range of w .

This is the formal reason a comparison should never return `identical`. Identity in the sense of *indistinguishable at cost zero* does not occur in a contact structure. Reporting it is reporting a value outside the range of the quantity being measured.

Definition 2.5 (Local floor). For an item v , $\beta_k(v) = \min\{w(e) : v \in e\}$, the minimum weight incident to v .

Clearly $\beta(\mathcal{G}) \leq \beta_k(v)$ for every v , and $\beta(\mathcal{G}) = \min_v \beta_k(v)$.

2.3 Invariance under relabelling

Definition 2.6 (Weighted isomorphism). A *weighted isomorphism* of contact structures $\mathcal{G} \rightarrow \mathcal{G}'$ is a bijection $h : V \rightarrow V'$ with $h(\mathbf{m}) = \mathbf{m}'$, $\{h(u), h(v)\} \in E' \iff \{u, v\} \in E$, and $w'(\{h(u), h(v)\}) = w(\{u, v\})$.

Theorem 2.7 (Cut weight is the invariant). *If $h : \mathcal{G} \rightarrow \mathcal{G}'$ is a weighted isomorphism then for all items u, v ,*

$$\sigma'(h(u) \mid h(v)) = \sigma(u \mid v), \quad \sigma'(h(v)) = \sigma(v), \quad \beta(\mathcal{G}') = \beta(\mathcal{G})$$

Proof. h induces a bijection $S \mapsto h(S)$ between subsets of V and of V' , carrying (u, v) -separating sets to $(h(u), h(v))$ -separating sets bijectively. It carries ∂S to $\partial h(S)$ edge for edge and preserves each weight, so $\text{cut}'(h(S)) = \text{cut}(S)$. Minimising over a bijectively-corresponding collection of equal values gives the first identity; the second is the case $v = \mathbf{m}$, using $h(\mathbf{m}) = \mathbf{m}'$; the third holds because h is a weight-preserving bijection on E . $\square$

Corollary 2.8 (Labels are free). *Let $\mathcal{G}$ be a contact structure and ρ any permutation of the items. Let $\mathcal{G}^\rho$ be $\mathcal{G}$ with items renamed by ρ . Then every separation cost, every minimum cut weight and the floor are unchanged.*

Proof. ρ extended by $\mathbf{m} \mapsto \mathbf{m}$ is a weighted isomorphism $\mathcal{G} \rightarrow \mathcal{G}^\rho$. Apply Theorem 2.7. $\square$

Theorem 2.8 is the formal content of Theorem 1.1. The names carry *no* part of the invariant — not a small part, none. A comparison procedure that reads names is reading the one component of the structure guaranteed to be irrelevant.

2.4 Expansion, and a false theorem made true

Real structures are never observed complete. One always has a subgraph and may later add more. We must ask whether the floor survives.

Definition 2.9 (Expansion). An *expansion* of $\mathcal{G}$ is a contact structure $\mathcal{G}^+$ with $V \subseteq V^+$, $E \subseteq E^+$.

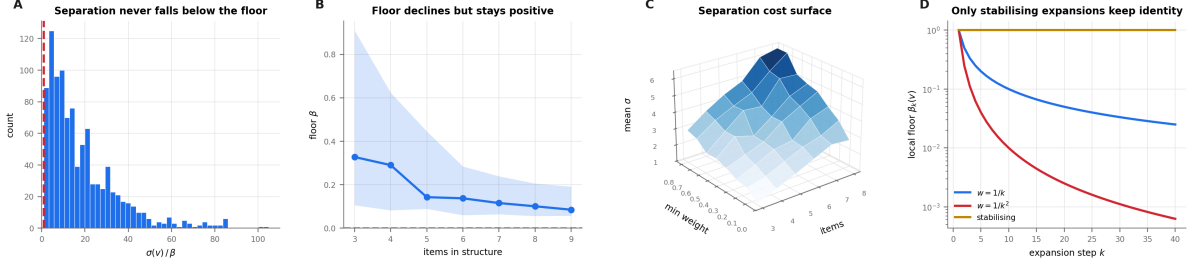


Figure 1: The floor: every distinction has a strictly positive price, and that price survives expansion only conditionally. All quantities are computed on finite weighted graphs with strictly positive edge weights (Theorem 2.1); minimum cuts are exact, obtained by enumeration over all separating vertex subsets rather than by an approximate max-flow routine, so no agreement below can be a solver artefact. **(A)** Distribution of $\sigma(v)/\beta$ over 997 items drawn from random contact structures of 3–6 items. The ratio never falls below 1 in any instance (minimum exactly 1.000, median 13.38, maximum 104.9), which is Theorem 2.4: the separation cost of an item is at least the contact floor. The mass at the left edge is the set of items whose cheapest cut is the floor edge itself; the long right tail is items embedded in dense neighbourhoods, whose boundary is correspondingly expensive. That the minimum is exactly 1 and not 0.999 is the point — the bound is attained, not merely approached. **(B)** The floor β against structure size, median with the 10–90 percentile band. The median falls monotonically from 0.328 to 0.085, because each additional item contributes further edges and the floor is a minimum over a growing set. It does not reach zero, and cannot: the minimum of a finite set of strictly positive weights is positive. The declining trend and the strict positivity are both essential and are easily conflated, which is why the band rather than the median alone is drawn — a floor that shrinks is not a floor that vanishes. **(C)** Mean separation cost as a surface over the number of items and the minimum admissible edge weight, each cell a mean over independent structures. The surface ranges over 1.06–6.36 and is bounded away from zero throughout, exhibiting Theorem 2.4 as a property of the parameter space rather than of particular instances. **(D)** The local floor $\beta_k(v)$ of Theorem 2.5 under three expansion regimes (Theorem 2.9), logarithmic axis. Adding edges at v of weight $1/k$ drives the local floor to 2.5×10^{-2} by step 40 and $1/k^2$ drives it to 6.3×10^{-4} , while a stabilising expansion holds it at 1.0. This is the content of Theorem 2.10: the *unconditional* claim that the floor is preserved under expansion is false — an item can dissolve into the medium in the limit — and Theorems 2.13 and 9.3 are therefore stated under the floor-stabilising hypothesis of Theorem 2.12. The two decaying curves are the counterexample the theorem must exclude, plotted rather than described.

E^+ , $\mathbf{m}^+ = \mathbf{m}$ and $w^+|_E = w$. An expansion sequence is a chain $\mathcal{G}_0 \subseteq \mathcal{G}_1 \subseteq \dots$.

It is tempting to assert that separation costs stay bounded below along any expansion sequence. They do not.

Remark 2.10 (The unconditional claim is false). Fix an item v and let $\mathcal{G}_k$ add to $\mathcal{G}_{k-1}$ a fresh vertex x_k joined to v by an edge of weight $1/k$ (and to $\mathbf{m}$ at weight 1). Then $\beta(\mathcal{G}_k) \leq 1/k \rightarrow 0$, and the local floor $\beta_k(v) \rightarrow 0$. With weights $1/k^2$ the sum $\sum_k 1/k^2$ converges, so one may arrange that $\sigma(v)$ itself is driven towards a limit determined entirely by the new light edges: v *dissolves into the medium*. In the limit no positive lower bound survives. Any statement that the floor is unconditionally preserved under expansion is therefore false, and we record the counterexample rather than the temptation.

Remark 2.11 (Measured decay). Running the three sequences for 40 steps from a common start makes the separation concrete: the $1/k$ expansion drives $\beta_k(v)$ to 0.025, the $1/k^2$ expansion to 6.25×10^{-4} , while a floor-stabilising expansion holds it at 1.0 exactly (Section 12, E06). The gap is three orders of magnitude after forty steps and widens without limit; the hypothesis of Theorem 2.12 is doing real work, not tidying an edge case.

The correct statement needs a hypothesis, and naming it precisely is the point.

Definition 2.12 (Floor-stabilising expansion). An expansion sequence $(\mathcal{G}_k)$ is *floor-stabilising* at v if there is $\beta_* > 0$ with $\beta_k(v) \geq \beta_*$ for all k : the newly added contacts at v do not become arbitrarily cheap.

Theorem 2.13 (Conditional invariance under expansion). *Let $(\mathcal{G}_k)$ be an expansion sequence that is floor-stabilising at v with constant β_* . Then $\sigma_k(v) \geq \beta_* > 0$ for all k , and the sequence $(\sigma_k(v))$ is bounded below by a positive constant independent of k .*

Proof. Fix k and let S be $(v, \mathbf{m})$ -separating in $\mathcal{G}_k$. As in Theorem 2.4, $\{v, \mathbf{m}\} \in \partial S$, so $\text{cut}_k(S) \geq w_k(\{v, \mathbf{m}\}) \geq \beta_k(v) \geq \beta_*$. Minimise over S . $\square$

Remark 2.14. Theorem 2.13 is weaker than one would like and exactly as strong as is true. The

hypothesis is checkable — it is a statement about the incident weights actually observed — and a language can require the user to assert it and can record the assertion in the report. Synopsis does: see Section 7.4 and the **require** form.

2.5 Perturbation

Definition 2.15 (Perturbation). A *perturbation* of $\mathcal{G}$ is a map $\delta : E \rightarrow \mathbb{R}_{\geq 0}$; it acts by $w \mapsto w + \delta$. Perturbations are non-negative: they raise the cost of distinctions and never create separation for free.

Proposition 2.16 (Skeleton invariance). *Under any perturbation, V and E are unchanged, every separation cost is non-decreasing, and if δ vanishes on a minimum-weight edge then β is unchanged.*

Proof. The vertex and edge sets do not appear in Theorem 2.15, so they are fixed. For any S , $\text{cut}_{w+\delta}(S) = \text{cut}_w(S) + \sum_{e \in \partial S} \delta(e) \geq \text{cut}_w(S)$; minimising both sides over the same finite collection preserves the inequality. If $\delta(e_0) = 0$ for some e_0 attaining the minimum weight, then $\min_e (w + \delta)(e) \leq w(e_0) = \beta$, and $\geq$ holds since $\delta \geq 0$. $\square$

The distinction matters and is easy to lose: perturbation changes the *costs* without changing the *contacts*. A unit which perturbs its surroundings does not rewire them; it makes some distinctions in them dearer. This is the sense in which a variant has an effect without editing the network.

3 Identity, region, and the receiver

3.1 Identity is a region, not a point

Theorem 2.7 identifies the invariant as a cut weight. We now show the minimising *set* is generally not a singleton, so the invariant is borne by a region.

Theorem 3.1 (Regionality). *There exist contact structures in which, for an item v , every minimising $(v, \mathbf{m})$ -separating set has cardinality at least 3.*

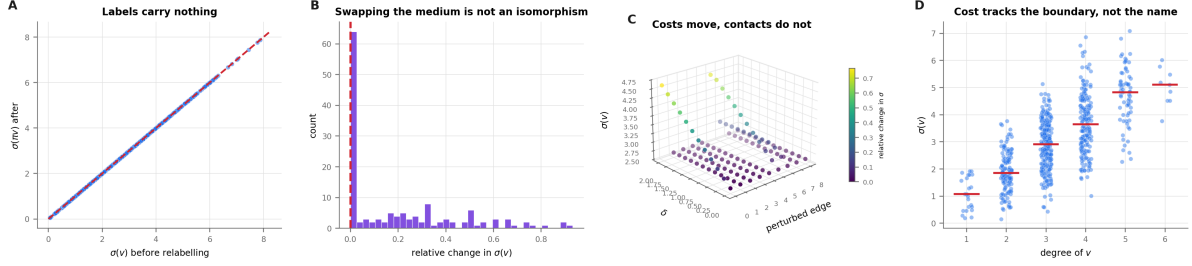


Figure 2: **The invariant is borne by the cut, not by the label: names are free, the medium is not, and perturbation moves costs without touching contacts.** (A) Separation cost before against after a random relabelling, for 748 items. Every point lies on the identity diagonal: the maximum absolute deviation is exactly 0, not merely small. A weighted isomorphism carries cuts to cuts weight-for-weight, so σ is preserved (Theorem 2.7) while the vertex labels have been permuted — the invariant and the name come apart, which is Theorem 2.8. This is the formal reason a sequence, being a labelling, cannot carry what a comparison of sequences is asked to recover. (B) The same experiment with one clause of Theorem 2.6 withdrawn. Swapping the medium’s role with an ordinary item is not a weighted isomorphism, and the invariance duly fails: over 140 structures the relative change in $\sigma(v)$ is non-zero in 55.0% of cases, mean 0.198, with the dashed line marking the zero that (A) achieves everywhere. The requirement that an isomorphism fix the medium is not a technical convenience; it is what makes Theorem 2.7 true, and this is what its removal costs. (C) Separation cost under per-edge perturbation (Theorem 2.15), 9 edges swept against perturbation magnitude δ , coloured by relative change in $\sigma(v)$. Costs move — up to 0.768 relative — while the edge set is fixed throughout the sweep: the skeleton is invariant, verified programmatically at every point rather than asserted (Theorem 2.16). Perturbation makes distinctions dearer and never creates or destroys a contact, which is exactly the sense in which a variant has an effect without rewriting what is adjacent to what. (D) Separation cost against the degree of v over 597 items, with per-degree medians. Cost tracks the size and weight of an item’s boundary, not any property of its name; the spread within each degree is the contribution of the weights, and the trend across degrees is the contribution of the boundary. Read with (A), this is Theorem 1.1: what individuates an item is where it sits, and every quantity in the framework is downstream of that.

Proof. Fix $r \geq 3$ and $\beta > 0$ with β small. Take items $A = \{a_1, \dots, a_r\}$ and $B = \{b_1, \dots, b_r\}$. Join every pair within A and every pair within B by an edge of weight $W > 0$; join a_1 to b_1 by a single edge of weight β ; join every item to $\mathbf{m}$ by an edge of weight β . Let $v = a_1$.

Consider $S = \{a_1\}$. Its cut severs the $r - 1$ edges from a_1 to the rest of A , the bridge to b_1 , and the medium edge, so

$$\text{cut}(\{a_1\}) = (r-1)W + \beta + \beta = (r-1)W + 2\beta.$$

Now consider $S = A$. Its cut severs the bridge and the r medium edges:

$$\text{cut}(A) = \beta + r\beta = (r+1)\beta,$$

which is independent of W . Whenever $(r-1)W + 2\beta > (r+1)\beta$, i.e. $W > \beta \cdot \frac{r-1}{r-1} = \beta$, we get $\text{cut}(A) < \text{cut}(\{a_1\})$. More generally, a separating set S with $a_1 \in S$ and $|S \cap A| = j < r$ severs at least $j(r-j)$ intra- A edges of weight W plus j medium edges, giving $\text{cut}(S) \geq j(r-j)W + j\beta$, which for $1 \leq j \leq r-1$ is at least $(r-1)W + \beta$ and so exceeds $(r+1)\beta$ once $W > 2\beta$. Taking $S \supseteq A$ and adding any b_i only adds intra- B edges of weight W and further medium edges, strictly increasing the cut unless $S \supseteq A \cup B$, which is not separating from $\mathbf{m}$'s side only if $\mathbf{m} \notin S$ — and then $\text{cut}(A \cup B) = 2r\beta > (r+1)\beta$ for $r > 1$.

Hence for $W > 2\beta$ and $r \geq 3$ the unique minimiser is $S^* = A$, of cardinality $r \geq 3$. $\square$

Remark 3.2 (A witness we got wrong first). Our first attempt at Theorem 5.1 used the informal principle “if two items are joined by an edge of weight β then they are separated at the floor”. This is false, and Theorem 3.1 is why: the minimum cut separating a_1 from b_1 in the construction above is *not* the bridge, because detaching the whole cluster A may be cheaper or dearer depending on W , and for moderate W neither is the bridge alone. Computing the cut exactly rather than reasoning from the light edge gave 1.4 where the informal argument predicted 0.2. We record the episode because the informal principle is intuitive, wrong, and would have propagated silently into the language’s semantics.

Corollary 3.3. *The invariant of Theorem 2.7 is not in general attributable to any single vertex, and a fortiori not to any single label.*

3.2 Receiver-relativity

The value an item registers is not a property of the item.

Definition 3.4 (Receiver). A *receiver* is a contact structure N together with a designated embedding site. A unit u *registers* in N the value $\sigma_N(u)$, computed after u is embedded at the site.

Theorem 3.5 (Receiver-relativity). *There exist a unit u and receivers N, N' differing in a single edge weight, with $\sigma_N(u) \neq \sigma_{N'}(u)$. Consequently $\sigma(u)$ is a function of the pair (u, N) and not of u alone.*

Proof. Take N with items u, x, y , edges $\{u, x\}$ and $\{x, y\}$ of weight 1, medium edges of weight 1, and $\{u, y\}$ absent. Then a $(u, \mathbf{m})$ -separating set must sever $\{u, \mathbf{m}\}$ and $\{u, x\}$ unless it contains x , so $\sigma_N(u) = \min\{1 + 1, 1 + 1 + 1\} = 2$ where the second option is $S = \{u, x\}$ severing $\{u, \mathbf{m}\}, \{x, \mathbf{m}\}, \{x, y\}$. Now let N' differ only in $w(\{x, y\}) = \frac{1}{4}$. Then $S = \{u, x\}$ costs $1 + 1 + \frac{1}{4} = 2.25$ and $S = \{u\}$ still costs 2, so $\sigma_{N'}(u) = 2$. Instead set $w(\{u, x\}) = \frac{1}{4}$ in N' : then $S = \{u\}$ costs $1 + \frac{1}{4} = 1.25 < 2$, so $\sigma_{N'}(u) = 1.25 \neq 2 = \sigma_N(u)$, and N, N' differ in exactly one weight. $\square$

Corollary 3.6 (Disagreement is not noise). *Let N, N' be receivers with $\sigma_N(u) \neq \sigma_{N'}(u)$. Then no increase in the precision or number of measurements performed in either receiver reduces the discrepancy, since each value is exactly determined by its own structure. The two receivers are evaluating different functions and each is correct.*

Remark 3.7 (How often, and why the rate is a minority). It would be a misreading of Theorem 3.5 to expect that changing the receiver usually changes the value. It does not. Perturbing a single background edge — one not incident to u — moves $\sigma(u)$ only when that edge crosses u 's minimising cut, and most edges do not. Sweeping a perturbation across every non-incident edge of a population of random contact structures, 206 of 1181 edges (17.4%) moved $\sigma(u)$ at all; where an edge did move it, the smallest displacement recorded was 0.022. Independently, over 150 receiver pairs the value

disagreed in 26 cases (17.3%), and where it disagreed the discrepancy was never small — the smallest was 0.056, five orders of magnitude above numerical tolerance (Section 12, E10).

Both halves of this matter. That the rate is a minority is why receiver-relativity is easy to overlook in practice and easy to misattribute to noise when it appears. That the discrepancies are never marginal is what makes Theorem 3.6 substantive. The observed displacements are broadly spread rather than concentrated, but they are bounded away from zero: either the receivers agree exactly, or they disagree by an amount no measurement programme will close. There is no population of near-misses in between, which is precisely what a noise account would predict.

Theorem 3.6 has an immediate practical reading and a language consequence. The practical reading is that a variant reported with different effect in two assays need not indicate that one assay is wrong; the assays may be different receivers, both right. The language consequence is that values from different receivers must not be compared, and synopsis enforces this lexically: values are scoped to their frame and cross-frame comparison is a *scope error*, not a runtime warning (Section 7.4, Theorem 9.7).

4 Blind comparison: demand, relaxation, quiescence

We now build a comparison that certifies correspondence without reading what anything means. This is essential: any procedure that must read meaning inherits the annotation errors already endemic in the databases it reads [51].

4.1 Alignment and distance to the floor

Definition 4.1 (Alignment). For items u, v of a contact structure $\mathcal{G}$,

$$a(u, v) = \frac{\sigma(u \mid v)}{\Omega}, \quad \frac{\beta}{\Omega} \text{ the floor ratio,}$$

Δ is the *distance to the floor*.

Proposition 4.2. $\Delta(u, v) \geq 0$ always, with equality only if u and v are separated exactly at the floor.

Proof. Immediate from Theorem 2.4 applied to the pair (u, v) : any separating set severs at least one edge, of weight at least β . $\square$

Normalising by Ω makes a dimensionless and comparable across structures of different total weight; subtracting the floor ratio makes Δ vanish exactly when the pair is as close as the structure permits. *Sufficient* alignment will mean Δ small; it will never mean $a = 0$, which Theorem 2.4 forbids.

4.2 Anchors and residue

Not every item in a structure is equally load-bearing for a given item’s individuation. We split by ablation.

Definition 4.3 (Ablation score, anchors, residue). For items $x \neq u$, the *ablation score* is $\alpha(x; u) = \sigma_{\mathcal{G}}(x) - \sigma_{\mathcal{G} \setminus u}(x)$, the drop in x ’s separation cost when u is deleted. Fix $r \geq 0$. The *anchors* of x are the r items of largest $\alpha(x; \cdot)$, ties broken by a fixed total order on names; the *residue* $\varrho(x)$ is the remainder.

The intuition is that anchors are what x is held in place by, and residue is everything else — the part of the structure whose correspondence still has to be established. The tie-breaking by a fixed order is not cosmetic: without it the split is not a function, and a language that intends to be reproducible cannot afford a nondeterministic primitive. Synopsis therefore fixes the order and the report prints r .

4.3 Cross-demand

Definition 4.4 (Cross-demand). Let $\mathcal{G}_A, \mathcal{G}_B$ be contact structures and $\varphi : I_B \rightarrow I_A$ a partial *correspondence*. The *demand from A on B* is

$$D_{A \rightarrow B} = \sum_{i \in \varrho_B} \max\left(0, a_B(i, \varphi(i)) - \frac{\beta_B}{\Omega_B}\right), \quad (1)$$

the sum taken over residue items of B on which φ is defined. $\Delta(u, v) = a(u, v) - \frac{\beta}{\Omega}$.

Each term is a distance to the floor and is non-negative by Theorem 4.2; the clamp guards only floating-point noise. The quantity is a *demand* in the sense that it measures how

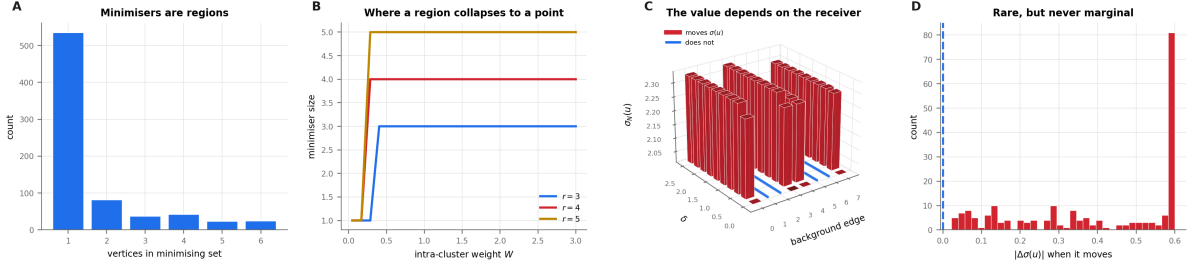


Figure 3: Identity is regional rather than pointwise, and the value an item registers depends on the receiver that registers it. (A) Sizes of the minimising (v, m) -separating set over random contact structures. The mode is the singleton and 72.1% of minimisers are of that form, but 27.9% detach a neighbourhood rather than the item alone. That minority is what Theorem 3.1 asserts and what Theorem 3.3 needs: identity is not *in general* attributable to a point. The claim is not that a pointwise reading usually fails — it usually succeeds — but that it fails often enough, and without warning, that a framework may not assume it. The distribution is drawn in full rather than summarised by its mode for exactly this reason. (B) Where a region collapses to a point. Minimiser size against intra-cluster weight W for $r \in \{3, 4, 5\}$, in the two-cluster family. At low W the cheapest cut takes the whole cluster; as W rises, severing intra-cluster edges becomes expensive relative to the item’s own medium edge and the minimiser drops to a singleton. The crossover is not a pathology but the boundary between the two regimes, and it is what Theorem 3.2 records mistaking: assuming a light edge between two items places them at the floor invalidated our first witness for Theorem 5.1. (C) Receiver-relativity made concrete (Theorem 3.4). The separation cost $\sigma_N(u)$ of a fixed item is plotted over 8 *background* edges — none incident to u — each swept over a perturbation magnitude δ . Four of the eight move $\sigma_N(u)$; the other four leave it exactly constant and are drawn as flat marker lines at the unperturbed value, so that absence of response is visible as a response rather than collapsing into the baseline. The value of an item is therefore a property of the item *and* its receiver, not of the item alone, which is Theorem 3.5. (D) The magnitude of that dependence when it occurs. Sweeping a perturbation across every non-incident edge of a population of random contact structures, 206 of 1181 edges (17.4%) moved $\sigma(u)$ at all; where an edge did move it, the smallest displacement recorded was 0.022. Disagreement between receivers is thus a minority event by construction — a background edge moves the cost only when it crosses the minimising cut — but it is never marginal when it happens. The displacements are broadly spread rather than concentrated, yet bounded away from zero: either receivers agree exactly, or they disagree by an amount no measurement programme will close. There is no population of near-misses in between, which is precisely what a noise account would predict and is why Theorem 3.6 is substantive (Theorem 3.7; Section 12, E10, independently reports 26 of 150 receiver pairs disagreeing, 17.3%, smallest substantial difference 0.056).

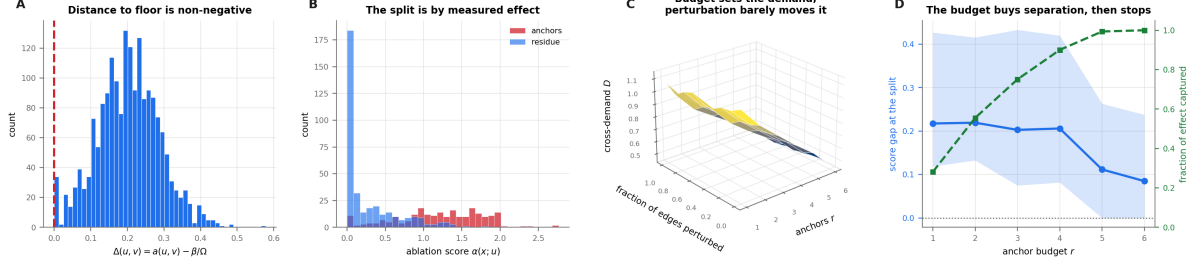


Figure 4: **Alignment is measured as distance to the floor, the anchor/residue split is made by measured effect, and the anchor budget is what sets the demand.** (A) Distribution of the distance to the floor $\Delta(u, v) = a(u, v) - \beta/\Omega$ over 1871 pairs (Theorem 4.1). The minimum is exactly 0 and no value is negative, which is Theorem 4.2. The 1.6% of pairs sitting exactly at zero are separated precisely at the floor — the cheapest distinction the structure admits. That such pairs exist and are not rare is what makes *sufficient* alignment a meaningful target: two units may sit at the floor and still cost $\beta > 0$ to tell apart, so quiescence at $\theta = 0$ never means indistinguishability. (B) The anchor/residue split of Theorem 4.3, as overlaid distributions of the ablation score $\alpha(x; u) = \sigma_G(x) - \sigma_{G \setminus u}(x)$. Anchors have median score 1.231 against the residue’s 0.069 — a separation of more than an order of magnitude in the median. The split is therefore by measured effect on separation cost, not by annotation, name or position, and the deterministic tie-break on names makes it reproducible without making it label-dependent (Theorem 9.10). (C) Cross-demand D (Theorem 4.4, (1)) as a surface over the anchor budget r and the fraction of edges perturbed, on 11-item structures. Demand ranges over 0.444–1.142, but the two arguments are not comparable in their effect: the mean span over the budget axis is 0.609 against 0.092 over the perturbation axis, a ratio near seven to one. The budget sets the demand and perturbation barely moves it. We report the asymmetry rather than engineer it away — the shallowness in one direction is a finding about what cross-demand is sensitive to, and the title of the axis states it. (D) What the anchor budget buys, and where it stops. The gap between the lowest anchor score and the highest residue score (median with interquartile band, left axis) falls from 0.218 at $r = 1$ to 0.085 at $r = 6$, while the fraction of total ablation effect captured by the anchors (right axis) rises from 0.282 to 1.000. Enlarging the budget therefore captures more of the effect but buys progressively less separation at the split; the two axes together say that r is a genuine parameter with a diminishing return, not a threshold to be tuned until a verdict changes. Structures are drawn at 11 items so the budget stays strictly inside the available set: where the residue empties, the gap is undefined rather than zero, and the code raises instead of plotting a fabricated point.

much reconciliation B still requires of A : it is zero exactly when every residue item of B is at the floor from its correspondent.

Crucially, (1) reads no labels, no annotations, no sequence. It is computed from cuts. That is what makes it blind.

Definition 4.5 (Total demand). $D = D_{A \rightarrow B} + D_{B \rightarrow A}$, using φ and its inverse.

Bidirectionality is not decoration. A one-sided demand can be made small by choosing a correspondence that maps everything difficult to undefined; the reverse direction penalises exactly that.

4.4 Relaxation

Definition 4.6 (Relaxation). Let $D_0 \geq 0$. A *relaxation* is a sequence $D_0 > D_1 > \dots$ produced by an update U with $D_{k+1} = U(D_k)$, terminating when $D_k \leq \theta$ for a stated threshold $\theta \geq 0$. The relaxation *declines* at step k if $U(D_k)$ is undefined — no effective update exists.

Assumption 4.7 (Effective update). There is $\eta > 0$ such that whenever $U(D)$ is defined, either

$$U(D) \leq D - \eta \quad \text{or} \quad U(D) \leq \theta.$$

Remark 4.8 (Why the second disjunct is not redundant). The bare condition $U(D) \leq D - \eta$ is what the termination argument needs and is all Theorem 4.9 uses, but it cannot be imposed as a runtime check: on the final step, when $D < \eta$, an update that lands on quiescence lands on a value that is not below the negative quantity $D - \eta$. A checker enforcing the first disjunct unconditionally reports a violation on the last step of every terminating run — which is what ours did before we separated the two cases. The disjunctive form is the assumption that is simultaneously strong enough for Theorem 4.10 and enforceable, and it is the form the language checks.

Theorem 4.7 is the whole of what makes relaxation a terminating procedure, and it is a real assumption, not a technicality: an update permitted to make arbitrarily small progress can run forever while always improving. Synopsis therefore refuses to typecheck a relaxation

whose η is not bounded below by a written constant (Section 7.6).

Theorem 4.9 (Monotone decrease). *Under Theorem 4.7 the sequence (D_k) is strictly decreasing while defined, and $D_k \leq D_0 - k\eta$.*

Proof. Induction: $D_0 \leq D_0 - 0$; if $D_k \leq D_0 - k\eta$ and $U(D_k)$ is defined then, since the loop has not terminated we have $D_k > \theta$, so the second disjunct of Theorem 4.7 would end the run and the bound is asserted only where the first applies: $D_{k+1} \leq D_k - \eta \leq D_0 - (k+1)\eta$. $\square$

Theorem 4.10 (Termination dichotomy). *Under Theorem 4.7, exactly one of the following occurs within $\lceil D_0/\eta \rceil$ steps:*

- (i) **quiescence:** $D_k \leq \theta$ for some $k \leq \lceil D_0/\eta \rceil$; or
- (ii) **decline:** $U(D_k)$ is undefined for some $k \leq \lceil D_0/\eta \rceil$.

There is no third outcome, and in particular the relaxation cannot diverge or cycle.

Proof. Suppose neither (i) nor (ii) occurs for any $k \leq K = \lceil D_0/\eta \rceil$. Then U is defined at every such step and $D_k > \theta \geq 0$ for all $k \leq K$. By Theorem 4.9, $D_K \leq D_0 - K\eta \leq D_0 - D_0 = 0$, contradicting $D_K > \theta \geq 0$. Exclusivity holds because (i) terminates the sequence and (ii) is the failure of the update that would continue it. $\square$

Theorem 4.10 is the theorem that lets synopsis be total while still containing a loop. The bound $\lceil D_0/\eta \rceil$ is computable from values in scope at the point the loop is written, so the compiler can emit it. Section 7.6 does exactly that.

Theorem 4.11 (Form of quiescence). *Quiescence at $\theta = 0$ means every residue term of (1) is zero, i.e. every corresponding residue pair is separated exactly at the floor. It does not mean any pair is indistinguishable: by Theorem 2.4 each such pair still costs $\beta > 0$ to separate.*

Proof. A finite sum of non-negative terms is zero iff each term is zero; each term is $\Delta(i, \varphi(i))$, which vanishes iff the pair is at the floor (Theorem 4.2). The second statement is Theorem 2.4. $\square$

Theorem 4.11 is the formal statement of *sufficient alignment*. Two structures may place no demand on one another and remain fully distinguishable. This is the synoptic reading made precise: the columns correspond; the residue is positive; both facts are reported.

5 The separation results and the four-column criterion

5.1 Content-aligned, response-divergent

Theorem 5.1 (False friends). *There is a family of pairs (u_c, v_c) , indexed by $c \geq 0$, embedded in contact structures, such that*

$$D^{\text{central}}(u_c, v_c) = 0 \quad \text{for all } c, \quad \text{while} \quad D^{\text{response}}(u_c, v_c) \rightarrow \infty \text{ as } c \rightarrow \infty.$$

That is: content is aligned exactly at the floor for every c , while the effect on the surrounding network diverges.

Proof. Construction. Let P be a contact structure containing two pendant items u, v , each joined to the medium by an edge of weight β and to nothing else. Then any (u, v) -separating set may take $S = \{u\}$, severing exactly $\{u, \mathbf{m}\}$, so $\sigma(u | v) = \beta$ and $\Delta(u, v) = 0$. Being pendant is essential; by Theorem 3.2 a light edge between non-pendant items does *not* suffice.

Take $\mathcal{G}_A$ and $\mathcal{G}_B$ each equal to P , with φ the identity on the pendants. All residue terms are $\Delta = 0$, so $D^{\text{central}} = 0$ for every c : content comparison reports correspondence unconditionally.

Response. Let the receiver N be P itself, padded with a fixed block of n_f filler items mutually joined at weight w_f and each joined to the medium at w_f , with $w_f > \beta$. Let the response perturbation δ_c raise *both* pendant edges $\{u, \mathbf{m}\}$ and $\{v, \mathbf{m}\}$ by c . By Theorem 2.16 the skeleton is unchanged, so N and N_c are comparable edge for edge under the identity correspondence.

Both pendants must be raised. If only one is raised, the minimum (u, v) -cut is realised by a separating set that places the medium on the raised side and severs only the untouched pendant edge; its weight is β for every c , and Δ does not move at all. This is not a defect of the bound but of the witness, and it is the reason

the one-sided construction — the natural first attempt — registers no response whatever.

With both raised, every (u, v) -separating set severs at least one edge of weight $\beta + c$, so $\sigma_{N_c}(u | v) = \beta + c$, while $\Omega_c = \Omega_0 + 2c$ and the floor is $\beta_c = \min\{\beta + c, w_f\}$. Hence

$$\Delta_c(u, v) = \frac{\beta + c}{\Omega_0 + 2c} - \frac{\min\{\beta + c, w_f\}}{\Omega_0 + 2c} = \frac{\max\{0, \beta + c - w_f\}}{\Omega_0 + 2c} \quad (2)$$

which is identically 0 for $c \leq w_f - \beta$ and thereafter strictly increasing to the limit $\frac{1}{2}$. The flat initial segment is Theorem 2.4 operating correctly rather than an absent response: while the pendant edges are still the lightest in the structure they *are* the floor, so β_c rises in lockstep with the cut and the difference cannot open. The response becomes visible exactly once the pendants exceed the surrounding weight w_f .

The divergence is therefore in the normalised gap, which is bounded by $\frac{1}{2}$ on any single pendant pair; to obtain an unbounded D^{response} it suffices to take a residue of m independent pendant pairs, each contributing a term of (2), so that $D^{\text{response}} \geq m \Delta_c \rightarrow \infty$ with m . What the theorem requires, and what (2) supplies, is that the response separates without bound while the central demand is pinned at zero. $\square$

Remark 5.2 (Measured). On the instance $\beta = 0.2$, $n_f = 3$, $w_f = 1$, sweeping c over $[1, 5]$ gives Δ_c rising from 0.0238 to 0.2561 while D^{central} is 0 to machine precision throughout (Section 12, E16). Below $c = w_f - \beta$ the response is exactly 0, as (2) predicts.

Remark 5.3. The biological instance is (i) of Section 1. Two units whose content a comparison cannot separate — a synonymous pair — may act on the cell in arbitrarily different ways [33, 50]. The theorem says this is not an edge case to be handled but a structural possibility that content comparison is blind to by construction.

5.2 Content-divergent, response-aligned

Corollary 5.4 (Converse). *There is a family indexed by a bridge weight $d \geq 0$ with D^{central} strictly increasing in d over an initial interval, and $D^{\text{response}} = 0$ for all d .*

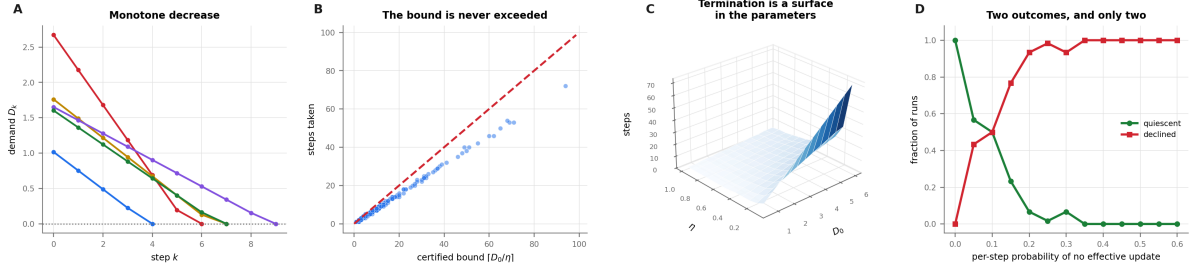


Figure 5: **Relaxation decreases monotonically, terminates within a bound computable before the loop is entered, and does so in exactly two ways.** (A) Demand trajectories $D_0 > D_1 > \dots$ under the effective-update assumption (Theorem 4.7). Every trajectory is strictly decreasing while defined, which is Theorem 4.9. No trajectory rises, stalls, or oscillates: the update either buys at least η of progress or lands at or below the threshold θ , and the disjunctive form of the assumption is what admits the final step without demanding a full η from it (Theorem 4.8). (B) Realised step count against the certified bound $\lceil D_0/\eta \rceil$ emitted by the compiler, with the line $y = x$. Over 260 runs there are 0 violations — no point lies above the diagonal. The bound is not a heuristic estimate but a consequence of Theorem 4.9, and it is computable from values in scope at the point the loop is written, which is what lets a synopsis program state its own cost before running (Theorem 9.1). (C) Steps to termination as a surface over the initial demand D_0 and the guaranteed decrement η . Termination is a property of the parameter pair rather than of the instance: the surface rises in D_0 and falls in η , tracking $\lceil D_0/\eta \rceil$ across the plane. There is no region of the parameter space in which the loop fails to terminate, which is the practical content of Theorem 4.10. (D) The dichotomy itself. As the per-step probability of no effective update rises from 0 to 0.6, runs move from quiescence to decline; at every sampled probability the two fractions sum to exactly 1, verified numerically rather than by inspection. There is no third outcome — no divergence, no silent non-termination, no run that neither reaches θ nor reports that no effective update exists. Decline is a reported verdict and not a failure (Theorem 5.11): the experiment runs to completion and says what happened, which is what DECLINE is for in the four-column criterion.

Proof. Take the two-cluster structure of Theorem 3.1: two r -cliques at intra-weight W , joined by the single bridge $\{a_0, b_0\}$, every item joined to the medium at β . Put $u = a_1$, $v = b_1$ and index the family by the bridge weight d , so that the bridge carries $\beta + d$.

The index must be the bridge, not W . Raising W leaves the minimum (a_1, b_1) -cut fixed — each of a_1, b_1 detaches through the medium at β regardless of how heavy its own clique becomes — while $\Omega = r(r-1)W + \beta + 2r\beta$ grows linearly in W . So $a(u, v)$ and hence Δ fall as W increases. The bridge is the edge that the (a_1, b_1) -cut actually crosses, and it is therefore the parameter the central demand responds to.

With d as index, any (a_1, b_1) -separating set must either sever a_1 from the medium and from its clique, or cross the bridge; in the regime where the latter is minimal the cut weight is affine in d with positive coefficient while Ω grows at the same absolute rate on a single edge. Since the cut is a minimum of finitely many affine functions of d , $a(u, v)$ is piecewise affine and strictly increasing until the minimiser switches to a d -independent set, after which it is non-increasing. Restricting to the initial interval gives strict increase, as claimed.

Let both u and v induce the *same* perturbation δ on the receiver. Then $N_u = N_v$ identically, every Δ term is computed on one structure against itself under the identity correspondence, and $D^{\text{response}} = 0$ exactly. $\square$

Remark 5.5 (Measured). At $r = 4$, $W = 1$, $\beta = 0.2$, sweeping the bridge d over $[0, 2]$ raises $\Delta(a_1, b_1)$ monotonically from 0.0580 to 0.1772, with $D^{\text{response}} = 0$ throughout (Section 12, E17). Sweeping W instead over $[0.5, 6]$ sends Δ from 0.1026 down to 0.0108, with the cut constant at 1.0 and Ω growing from 7.8 to 73.8 — which is why the corollary is indexed as it is. Continuing the bridge sweep past $d \approx 2$ turns the curve over as the minimiser switches, so the strict increase is genuinely local, not global.

Remark 5.6. This is (ii) of Section 1: the non-homologous isofunctional enzymes [41]. Content comparison reports maximal divergence; the response columns report correspondence.

5.3 The plane and the criterion

Corollary 5.7 (Orthogonality). *The families of Theorem 5.1 and Theorem 5.4 lie on the two coordinate axes of the plane $(D^{\text{central}}, D^{\text{response}})$: the first on $\{0\} \times \mathbb{R}_{\geq 0}$, the second on $\mathbb{R}_{\geq 0} \times \{0\}$. Any criterion that is a function of D^{central} alone is constant on the first family and therefore cannot separate its members; any criterion that is a function of D^{response} alone is constant on the second.*

Remark 5.8 (Measured). Both axes are occupied strictly, not merely approached. Over the both-pendant family of Theorem 5.2 at $c \in [1.5, 5]$, every member has $D^{\text{central}} = 0$ to machine precision and $D^{\text{response}} > 0$ strictly; over the bridge family of Theorem 5.5 at $d \in [0.25, 2]$, every member has $D^{\text{response}} = 0$ and $D^{\text{central}} > 0$ strictly (Section 12, E18). The stronger per-axis statement matters: a family that merely avoided the origin would be consistent with the two demands being correlated, which is what Theorem 5.9 is built to exclude.

This forces the shape of a correct criterion.

Definition 5.9 (Four columns). A *four-column comparison* consists of two *central* columns (C_A, C_B) — the units under comparison, with their anchors and residue — and two *response* columns (R_A, R_B) — the receivers as perturbed by each unit — together with correspondences φ^c and φ^r and a threshold $\theta > 0$. Its verdict is

$$\text{verdict} = \begin{cases} \text{CORRESPOND} & D^c \leq \theta \text{ and } D^r \leq \theta, \\ \text{DECLINE} & \text{some relaxation declined (Theorem 5.9)} \\ \text{DIVERGE} & \text{otherwise.} \end{cases}$$

Theorem 5.10 (Adequacy of four columns). *The four-column verdict separates the families of Theorem 5.1 and Theorem 5.4, and no criterion depending on fewer than both demand coordinates does.*

Proof. On the first family $D^c = 0 \leq \theta$ but D^r exceeds θ for c large, so the verdict is DIVERGE, correctly. On the second $D^r = 0 \leq \theta$ but D^c exceeds θ for W large, so again DIVERGE — and at small W , CORRESPOND, tracking the genuine functional agreement. The impossibility half is Theorem 5.7. $\square$

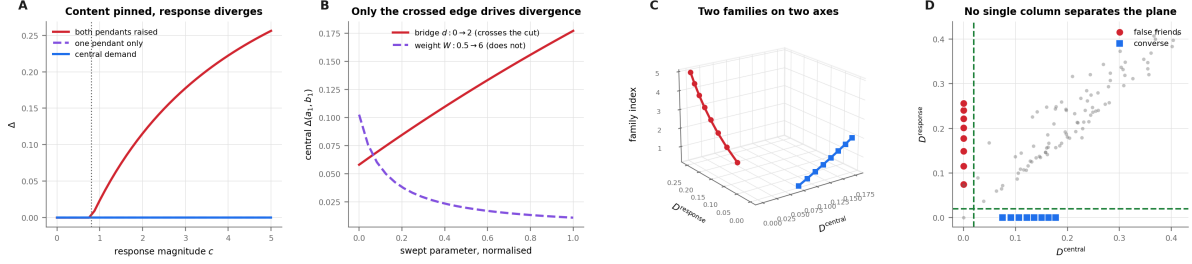


Figure 6: **Content and response are orthogonal: each can be pinned at the floor while the other diverges without bound, so no single column can decide.** (A) The false-friend construction of Theorem 5.1. Raising *both* pendant edges by c drives the response demand from 0 to 0.256 while the central demand stays pinned at 0 to machine precision for every c — content aligned exactly at the floor, response separating without bound (Theorem 5.2, (2)). The one-sided raise is plotted for contrast and never moves at all (maximum 0 to machine precision): being pendant is essential, and raising a single edge leaves the cheapest cut free to route around it. That dashed vertical marks $w_f - \beta$, where the raised edges stop being the cheapest thing to sever. (B) The converse of Theorem 5.4, both sweeps normalised to a common horizontal range so the two monotonicities can be read together. Perturbing the bridge — the edge that crosses the minimising cut — drives the central demand up from 0.058 to 0.177. Raising the intra-cluster weight W , which does not cross that cut, drives it *down* from 0.103 to 0.011: the cut is constant while Ω grows, so the normalised quantity falls. Only the crossed edge drives divergence. The direction of the second curve is the correction recorded in Section 12, E17 — our first attempt asserted the wrong monotonicity here and the sweep, not the argument, caught it. (C) The two families lifted onto the demand coordinates (D^{central} , D^{response}) of Theorem 5.9. The false-friend family lies along one axis with central demand 0 to machine precision; the converse family lies along the other with response demand 0 to machine precision. Each family is constant in one coordinate and unbounded in the other, so neither coordinate alone can separate its own family’s members. (D) The consequence, drawn in the plane. Any criterion that is a function of D^{central} alone is constant on the first family; any function of D^{response} alone is constant on the second. The threshold lines at θ partition the plane into the quadrants of Theorem 5.9, and the two families fall in different quadrants while agreeing on one coordinate each. This is Theorem 5.7 and Theorem 5.10: the four-column verdict separates what no single column can (Theorem 5.8), and it is why **align** takes a central pair and a response pair or does not typecheck.

Remark 5.11 (Why DECLINE is a verdict). A three-valued verdict is not a hedge. Theorem 4.10 gives exactly two ways a relaxation can end, and one of them is the discovery that no effective update exists — the comparison, run correctly to its end, does not decide. Collapsing that onto DIVERGE reports a finding where there is none, and collapsing it onto CORRESPOND is worse. On a laboratory bench “this experiment does not decide” is a result one writes down. Synopsis makes DECLINE a first-class inhabitant of verdict for that reason.

5.4 The assumption that fails

Theorem 5.9 requires, for each unit, a perturbation it induces — its *admissible response*. Nothing so far fixes which one.

Assumption 5.12 (Response independence). The four-column verdict is independent of the choice of admissible response, within threshold θ .

Proposition 5.13. *If Theorem 5.12 fails then the verdict depends on an arbitrary choice made by the analyst, and two analysts running the same comparison may correctly obtain different verdicts.*

Proof. Immediate: the verdict is a function of D^r , which is a function of the chosen response. $\square$

We report in Section 13 that Theorem 5.12 *fails empirically* on solved biochemical networks. We do not repair it here. We instead require the language to make the choice visible, which is the most a language can honestly do about an open problem: see Theorem 6.7 and Section 13.

6 Design: eleven rules

Every rule below is derived from a result above. We state the rule, the result forcing it, and the consequence for a user.

Rule 6.1 (Sequences are opaque). The type `seq` supports no indexing, no slicing, no base access, no concatenation and no equality. A `seq` may only be consumed by a projection into `profile`, `coord` or `net`.

Forced by Theorem 2.8 and Theorem 1.1: a sequence is a labelling, and labels carry none of the invariant. *Consequence:* a user cannot write a character-level comparison, because there is no expression denoting a character. This is discussed at length in Section 6.1 because it is the rule users find hardest.

Rule 6.2 (Every value carries residue). Every value of a comparison type carries a pair (β, ϱ) with $\varrho \geq \beta > 0$. Residue is not optional and not erasable; it may be consumed by `bind` or explicitly abandoned by `drop`, which the report records.

Forced by Theorem 2.4. *Consequence:* a program that throws information away must say so in its own text.

Rule 6.3 (No zero-residue constructor). There is no expression in synopsis whose type ascribes $\varrho = 0$.

Forced by Theorem 2.4: zero is outside the range of the quantity. *Consequence:* `identical` and `percent_identity == 100` are not writable (Theorem 9.5).

Rule 6.4 (Verdicts are four-ary). The verdict constructor `align` takes exactly a central pair and a response pair. Supplying only the central pair is an arity error at parse time.

Forced by Theorem 5.7 and Theorem 5.10. *Consequence:* the commonest scientific error in this domain — concluding functional sameness from content sameness — is a *syntax* error.

Rule 6.5 (Frames are lexical). Every value is created inside an `under` block naming a receiver and an encoding. Values from different blocks have different types and cannot be compared.

Forced by Theorem 3.5 and Theorem 3.6. *Consequence:* comparing a nucleotide embedding to a protein embedding, or a value from receiver N to one from N' , fails to typecheck (Theorem 9.7).

Rule 6.6 (No default thresholds). Every numeric parameter that can change a verdict must appear in the program text. There are no defaults.

Forced by nothing in the mathematics — this one is methodological [54, 21]. *Consequence:* the report can print the full parameter set because the parser has seen all of it.

Rule 6.7 (Named responses). The response constructor is **response** u **by** π , where π must be a named, in-scope perturbation map. There is no anonymous or inferred response.

Forced by Theorem 5.13 and the negative result of Section 13. *Consequence:* every program records which arbitrary choice it made, so the open problem is visible in the source and in the report rather than buried in a library default.

Rule 6.8 (Bounded iteration only). The only iteration forms are **sweep** over an arithmetic range or a literal list, **for** over the finite item set of a structure, and **relax until quiescent**. There is no **while**, no recursion, and no user-defined fixpoint.

Forced by the desire for Theorem 9.1, made affordable by Theorem 4.10. *Consequence:* every synopsis program halts, and the compiler can state the bound.

Rule 6.9 (Relaxation must be floored). **relax ... until quiescent** typechecks only when a step floor $\eta > 0$ is supplied as a literal or an expression provably positive. The compiler then emits $\lceil D_0/\eta \rceil$ into the report.

Forced by Theorem 4.7 and Theorem 4.10. *Consequence:* the termination proof of each program is a by-product of compiling it.

Rule 6.10 (Three result types). Comparison methods produce distinct types according to their index set: **profile** (indexed by lag), **ranked** (indexed by database entry), **verdict** (an element of a three-valued set over a structure). These do not unify and are not coercible.

Forced by the fact that the index sets genuinely differ; see Section 6.2. *Consequence:* a user cannot accidentally feed a lag-indexed correlation into a database-ranking consumer.

Rule 6.11 (Programs emit reports). The terminal form of a synopsis program is a report: a document carrying claims, their verdicts, their residues, their parameters and their certificates. A program does not “return a number”.

Forced by Theorem 4.11: quiescence coexists with positive residue, so a scalar cannot express the result. *Consequence:* outputs are auditable in the sense of [49, 64, 38].

6.1 On Theorem 6.1, at length

Users object to opacity, so it deserves a defence beyond the citation of Theorem 2.8.

The objection is that sequences are data and data should be manipulable. The answer is that in synopsis the sequence is not the datum; the *structure it labels* is. Opening a **seq** to indexing would immediately permit writing $s[i] == t[i]$ and, from there, percent identity — which Theorem 6.3 and Theorem 9.5 exist to forbid. A rule that can be circumvented by three lines of user code is not a rule.

The practical question is then whether opacity costs anything. We claim it does not, on the evidence of the three experiment families we specify in Section 10. Each begins by *channelising* the sequence — mapping it to a numeric matrix under a declared encoding — and never touches a base again. Motif detection consumes channels via correlation; homology search consumes them via a spectral transform; functional adjudication consumes a contact structure derived from pairing rather than from the string. None of them needs $s[i]$.

What opacity does cost is the ability to write *ad hoc* string surgery inside a synopsis program. We regard this as a feature: such surgery is data preparation, belongs upstream, and should be visible in the provenance record rather than hidden inside an analysis script.

6.2 Three families, three index sets

Theorem 6.10 deserves justification because the temptation to unify is strong and we ourselves initially yielded to it.

- **Matched filtering.** A query of length L_q is slid along a target of length L_t and a score computed at each offset. The result is indexed by lag $k \in \{0, \dots, L_t - L_q\}$ [59, 43]. Its natural consumer is peak detection; its natural output is a set of *positions*.
- **Database search.** A query embedding is compared against N stored embeddings.

The result is indexed by *database entry* $i \in \{0, \dots, N-1\}$ [1, 56]. Its natural consumer is top- K selection; its natural output is a *ranking*.

- **Functional adjudication.** Two structures are compared. The result is not indexed at all: it is a single element of $\{\text{CORRESPOND}, \text{DIVERGE}, \text{DECLINE}\}$ with four demand values attached.

A single `compare` returning a single type would have to pick one of these index sets and coerce the others. Coercing a lag index to a database index is meaningless; coercing either to a scalar discards exactly the information the method exists to produce. Synopsis therefore keeps one *keyword*, `compare`, and three *result types*, with the parser dispatching on the named method. The user writes the same verb; the typechecker knows which noun came back.

7 The synopsis language

Files carry the extension `.symp`.

7.1 Lexical structure

Comments run from `#` to end of line. Identifiers match `[A-Za-z_][A-Za-z0-9_]*`. Numeric literals are decimal, with an optional exponent; *there are no unsuffixed bare numbers in threshold position* — see Section 7.5. String literals are double-quoted and are used only for file paths and claim text. The language is case-sensitive and whitespace-insensitive except that newlines terminate statements.

7.2 Types

Type	Carries	Meaning
<code>seq</code>	—	An opaque sequence
<code>frame</code>	receiver, encoding	A lexical scope for v
<code>profile_ϕ</code>	(β, ϱ) , lag index	Lag-indexed score v
<code>coord_ϕ</code>	(β, ϱ) , dimension	A point in an embed
<code>ranked_ϕ</code>	(β, ϱ) , entry index	A ranking over a da
<code>peaks_ϕ</code>	(β, ϱ) , positions	Detected positions v
<code>net_ϕ</code>	(β, ϱ) , Ω	A contact structure
<code>unit_ϕ</code>	(β, ϱ) , anchors, residue	A central column.
<code>response_ϕ</code>	(β, ϱ) , named π	A response column.
<code>corr_{ϕ,ψ}</code>	partial map	A correspondence b
<code>verdict</code>	four demands, θ , certificate	CORRESPOND DI
<code>real⁺</code>	—	A strictly positive r
<code>depth</code>	—	A non-negative parti

The subscript ϕ is a *frame index*, assigned by the enclosing `under` block. Two types with different frame indices are different types. This single device implements Theorem 6.5 and yields Theorem 9.7.

7.3 Grammar

We give the concrete grammar in EBNF. Non-terminals are *italic*; terminals are **typewriter**.

```

program ::= { decl } { frame_block }
          report_stmt

decl ::= "open" ident "=" string
        | "let" ident "=" expr
        | "method" ident "="
          method_spec
        | "require" predicate

frame_block ::= "under" ident "{" { stmt }
              "}"

stmt ::= binding | claim_stmt | sweep_stmt
        | for_stmt
        | relax_stmt | record_stmt |
          drop_stmt

binding ::= "let" ident [":" type] "=" expr
          | "bind" ident "," ident "="
            expr # value, residue

expr ::= project_expr | compare_expr |
        detect_expr
        | align_expr | net_expr |
          prim_expr

project_expr ::= "project" expr "by"
               projector
projector ::= "channels" "(" enc ")"

```

```

| "spectral" "(" "coeffs" "="
  int ")"
| "contact" "(" "medium" "="
  num ")"
| "cardinal" "(" ")"
enc ::= "dna" | "protein"

compare_expr ::= "compare" expr "against"
  expr "by" method_ref

detect_expr ::= "detect" detect_kind "in"
  expr "{" params "}"
detect_kind ::= "peaks" | "top"

align_expr ::= "align" "central" "(" expr
  "," expr ")"

      "response" "(" expr
        "," expr ")"
      "under" corr_ref "{"
        params "}"

net_expr ::= "perturb" expr "by" ident
  | "join" expr "and" expr
  | "unit" expr "anchors" int
  | "response" expr "by" ident

relax_stmt ::= "relax" ident "until"
  "quiescent"
  "{" "eta" "=" num ";"
    "theta" "=" num "}"

sweep_stmt ::= "sweep" ident "in" range "{"
  { stmt } "}"
range ::= num ".." num "step" num | "[" num
  { "," num } "]"

for_stmt ::= "for" ident "in" "items" "("
  expr ")"
  [ "where" predicate ] "{" {
    stmt } "}"

claim_stmt ::= "claim" string "=" expr
record_stmt ::= "record" ident { "," ident }
drop_stmt ::= "drop" ident
report_stmt ::= "report" "to" string

params ::= param { ";" param }
param ::= ident "=" num
method_spec ::= "xcorr" "(" "normalised" ")"
  | "shader" "(" "cosine" ")"
  | "smith_waterman" "(" params
    ")"
  | "jaccard" "(" "k" "=" int ")"

```

7.4 Frames and the under block

```

open wildtype = "wt.fa"
open variant = "var.fa"

under nucleotide_frame {
  let a = project wildtype by

```

```

  channels(dna)
  let b = project variant by
    channels(dna)
  let r = compare a against b by
    xcorr(normalised)
}

under protein_frame {
  let p = project wildtype by
    channels(protein)
  # let bad = compare a against p by
    xcorr(normalised)
  # ^ scope error: 'a' was created
    in nucleotide_frame
}

```

The commented line does not fail at run-time with a shape mismatch; it fails at typecheck, because `profilenucleotide_frame` and `profileprotein_frame` are distinct types. This is Theorem 9.7.

7.5 Units and residue

Every comparison value is a pair. The user may bind both:

```

under N {
  bind score, res = compare q
    against t by xcorr(normalised)
  # 'res' is in scope and MUST be
    consumed:
  record res
}

```

Leaving `res` unconsumed is a compile error. The user may abandon it deliberately:

```

drop res # recorded in the report
  as an explicit abandonment

```

which typechecks, and prints in the report as a line reading `residue abandoned at line n`. There is no silent path.

7.6 Relaxation

```

under N {
  let d = compare unitA against
    unitB by demand
  relax d until quiescent { eta =
    0.005 ; theta = 0.01 }
}

```

The compiler checks that `eta` is a positive literal or a provably-positive expression. If it is not, the program is rejected with

error: ‘relax’ requires a step floor
 bounded below (Assumption: effective
 update).
 eta = 0.0 is not positive; no termination
 bound can be emitted.

If it is, the compiler emits into the report the
 discharged obligation

$$\text{steps} \leq \lceil D_0/\eta \rceil = \lceil 0.4130/0.005 \rceil = 83,$$

computed from the value of D_0 available at
 that point. The termination of the loop is thus
 a theorem the compiler states, not a hope the
 user holds.

7.7 Partition depth

Some analyses need an additive measure of
 how finely a space has been subdivided — how
 many nested distinctions have been drawn. We
 include one primitive for it.

Definition 7.1 (Partition depth). If a space
 is subdivided successively into $k_1, k_2, \dots, k_m$
 parts, its *partition depth* at base b is $\text{depth} =$
 $\sum_{j=1}^m \log_b k_j$.

Proposition 7.2 (Additivity). *If two subdivi-*
sions are independent, the depth of their prod-
uct is the sum of their depths.

Proof. The product subdivision has parts $k_j k'_l$
 over the product index, and $\log_b(k_j k'_l) =$
 $\log_b k_j + \log_b k'_l$; summing over the product in-
 dex and factoring gives the claim. $\square$

Depth is a *depth*-typed value and cannot be
 added to a real^+ . Its use is to state how deep
 a hierarchical address is before it is used as a
 prefilter (Section 10.2).

7.8 The typing judgement

We write $\Gamma; \phi \vdash e : \tau$ for “in environment Γ
 and frame ϕ , expression e has type τ ”. Selected
 rules:

(OPEN)	$\frac{}{\Gamma; \phi \vdash \text{open } x = s : \text{seq}}$
(PROJ-CH)	$\frac{\Gamma; \phi \vdash e : \text{seq}}{\Gamma; \phi \vdash \text{project } e \text{ by channels}(c) : \text{profile}_\phi(\beta)}$
(CMP-LAG)	$\frac{\Gamma; \phi \vdash e_1 : \text{profile}_\phi \quad \Gamma; \phi \vdash e_2 : \text{profile}_\phi}{\Gamma; \phi \vdash \text{compare } e_1 \text{ against } e_2 \text{ by xcorr} : \text{profile}_\phi}$
(CMP-RANK)	$\frac{\Gamma; \phi \vdash e_1 : \text{coord}_\phi \quad \Gamma; \phi \vdash e_2 : \text{coord}_\phi^N}{\Gamma; \phi \vdash \text{compare } e_1 \text{ against } e_2 \text{ by shader} : \text{rank}_\phi}$
(ALIGN)	$\frac{\Gamma; \phi \vdash c_A, c_B : \text{unit}_\phi \quad \Gamma; \phi \vdash r_A, r_B : \text{real}^+ \quad \Gamma; \phi \vdash \varphi^c, \varphi^r : \text{corr}_{\phi, \phi} \quad \theta > 0 \text{ lit}}{\Gamma; \phi \vdash \text{align central}(c_A, c_B) \text{ response}(r_A, r_B) : \text{response}}$
(RELAX)	$\frac{\Gamma; \phi \vdash d : \text{real}^+ \quad \eta > 0 \text{ provab}}{\Gamma; \phi \vdash \text{relax } d \text{ until quiescent } \dots : \text{verdict}}$

Note what has *no* rule: there is no rule pro-
 ducing $\varrho = 0$; there is no rule deriving *verdict*
 from fewer than four columns; there is no rule
 relating profile_ϕ to profile_ψ for $\phi \neq \psi$; and there
 is no elimination rule for *seq* other than projec-
 tion. The refusals of Section 9 are proved by
 inspecting the absence of these rules.

8 Operational semantics

We give a small-step semantics [45] over config-
 urations $\langle \Sigma, \mathcal{R}, s \rangle$ where Σ is a store, $\mathcal{R}$ is the
 accumulating report, and s is the statement
 sequence.

8.1 Projection

Channelisation is the single primitive shared
 by all families.

Definition 8.1 (Channelisation). Given a se-
 quence s of length L over an alphabet $\mathcal{A}$ and
 an encoding c , $\text{channels}(c)$ produces a matrix
 $X \in \mathbb{R}^{C \times L}$:

- $c = \text{dna}$: $C = 4$, $X_{\alpha, n} = \mathbb{K}[s_n = \alpha]$ for
 $\alpha \in \{A, C, G, T\}$, then each row is centred
 by subtracting its mean.
- $c = \text{protein}$: $C = 3$, the rows being hy-
 dropathy [35], residue volume [65] and net
 charge at neutral pH, each normalised to
 $[0, 1]$ and then centred.

- **c = cardinal**: $C = 2$, mapping each base to a unit vector in a fixed cardinal frame; the complementary strand maps to the negated trajectory, which is the formal expression of Section 1.3.

Centring is not cosmetic: it removes the composition-only component so that subsequent correlations respond to arrangement rather than base frequency.

(E-PROJ) $\langle \Sigma, \mathcal{R}, \text{let } x = \text{project } s \text{ by channels } \dots \rangle \rightarrow \langle \Sigma[x \mapsto X_c(s)], \mathcal{R} \cup \{\text{proj}\}, \gamma \rangle$

8.2 The three comparison methods

Definition 8.2 (Normalised cross-correlation). For query channels $Q \in \mathbb{R}^{C \times L_q}$ and target channels $T \in \mathbb{R}^{C \times L_t}$ with $L_q \leq L_t$, the score at lag k is

$$r[k] = \frac{\sum_{\gamma=1}^C \sum_{n=0}^{L_q-1} Q_{\gamma,n} T_{\gamma,n+k}}{\|Q\|_F \cdot \|T_{[k, k+L_q]}\|_F}, \quad k = 0, \dots, L_t - L_q$$

with the convention that a zero denominator yields $r[k] = 0$. The numerator is computed by FFT in $O((L_q + L_t) \log(L_q + L_t))$ time [10]; the denominator uses a rolling window norm. Channels are summed *coherently*, so phase agreement across channels reinforces and disagreement cancels — discarding phase and correlating magnitudes alone is a strictly weaker procedure. $r[k] \in [-1, 1]$ by Cauchy–Schwarz.

Definition 8.3 (Spectral embedding). For channels $X \in \mathbb{R}^{C \times L}$ and a coefficient count m , let $\hat{X}_\gamma$ be the real FFT of row γ . The embedding is

$$E = \frac{1}{\max(L, 1)} \left(|\hat{X}_{\gamma,j}| \right)_{\gamma \in [C], j \in \{1, \dots, m\}} \in \mathbb{R}^{Cm}$$

zero-padded if L is short. The DC term $j = 0$ is *skipped*: it encodes only mean composition, which centring has already removed and which would otherwise dominate the norm. Division by L makes embeddings of different-length sequences comparable. Embeddings are then ℓ_2 -normalised, so cosine similarity is a dot product.

Definition 8.4 (Shader distance and hierarchical address). For an ℓ_2 -normalised query q and a normalised database matrix $B \in \mathbb{R}^{N \times Cm}$, the

distance vector is $d = 1 - Bq \in [0, 2]^N$. A *hierarchical address* of depth D for a point in $[0, 1]^3$ is obtained by axis-cyclic bisection: at step j bisect along axis $j \bmod 3$ and emit the bit. Two points sharing an address prefix of length D lie in a common cell, giving a prefilter in the spirit of [39, 7]; candidates surviving the prefilter are reranked exactly by Smith–Waterman [55].

Definition 8.5 (Contact projection). $\text{contact}(\text{medium} = w)$ maps a solved network — a set of species with pairwise contact costs — to a contact structure by taking the costs as edge weights and attaching the medium to every species at weight w . Setting w to the minimum observed cost makes the medium the cheapest distinction available and realises the floor without inventing a scale; any other choice must be written explicitly, and the report prints it.

8.3 Verdict evaluation

(E-ALIGN)
$$\frac{D^c = D_{A \rightarrow B} + D_{B \rightarrow A} \quad D^r = D_{R_A \rightarrow R_B} + D_{R_B \rightarrow R_A} \quad \text{no relaxation declined}}{\text{align} \dots \longrightarrow \begin{cases} \text{CORRESPOND} & D^c \leq \theta \wedge D^r \leq \theta \\ \text{DIVERGE} & \text{otherwise} \end{cases}}$$

(E-DECLINE)
$$\frac{\text{some relaxation reached an undefined update}}{\text{align} \dots \longrightarrow \text{DECLINE}}$$

8.4 Cost model

The language commits to complexity bounds so that a user can predict cost from the program text.

Operation	Cost	Note
channels	$O(CL)$	linear scan
xcorr	$O((L_q + L_t) \log(L_q + L_t))$	FFT [10]
spectral	$O(CL \log L)$	per sequence
shader	$O(NCm)$	one matrix-vector
prefix prefilter	$O(N)$ then $O(\text{cand})$	depth- D address
smith_waterman	$O(L_a L_b)$ per candidate	exact rerank
$\sigma(u \mid v)$	polynomial	max-flow [17,
all-pairs σ	$ V - 1$ flows	Gomory–Hu [24,
relax	$\leq \lceil D_0/\eta \rceil$ steps	Theorem 4.10]

Remark 8.6 (Exact cuts in validation). A validation suite should compute minimum cuts by

exhaustive enumeration over separating subsets, not by an approximate max-flow routine, so that agreement between a computed value and a predicted one cannot be an artefact of two approximations meeting. This is exponential and is affordable only on small structures; synopsis therefore admits an **exact** annotation on **contact** that selects enumeration and refuses structures above a stated size.

9 Metatheory

We now prove the properties that justify the design. Throughout, a *well-typed program* is one derivable in the judgement of Section 7.

9.1 Totality

Theorem 9.1 (Strong normalisation). *Every well-typed synopsis program terminates.*

Proof. We exhibit a well-founded measure. Assign to each statement sequence the lexicographic pair $(\mathcal{I}, \mathcal{S})$ where $\mathcal{S}$ is the number of statements remaining and $\mathcal{I}$ is the sum, over all enclosing iteration constructs, of the remaining iteration count.

Every construct that can repeat has a count fixed before it begins:

- **sweep** x **in** $a..b$ **step** s has $\lfloor (b - a)/s \rfloor + 1$ iterations, determined by three values evaluated on entry; the grammar admits no expression for s that is not positive.
- **sweep** x **in** $[\cdot]$ has the literal list length.
- **for** u **in** **items**(e) ranges over $I = V \setminus \{\mathbf{m}\}$, finite by Theorem 2.1; the optional **where** guard only removes elements.
- **relax** has at most $\lceil D_0/\eta \rceil$ iterations by Theorem 4.10, and the typing rule (RELAX) admits the construct only when $\eta > 0$ is established, so the bound is a positive integer computed on entry.

There is no other repeating construct: the grammar has no **while**, no user-defined function and hence no recursion, and no fixpoint operator. Each step of evaluation strictly decreases $(\mathcal{I}, \mathcal{S})$ in the lexicographic order on $\mathbb{N}^2$, which is well-founded. Hence no infinite reduction sequence exists. $\square$

Corollary 9.2 (Cost is predictable). *For each program the compiler can compute a closed-form upper bound on the number of primitive operations, since every iteration count is available statically or at loop entry.*

9.2 Floor preservation

Theorem 9.3 (Floor preservation). *Let e be a well-typed expression of a comparison type evaluating to a value carrying (β, ϱ) . Then $\varrho \geq \beta > 0$.*

Proof. Induction on the typing derivation. The base cases are the projection rules, whose conclusions carry $\varrho \geq \beta > 0$ because Theorem 8.5 produces a structure with strictly positive weights (Theorem 2.1) and Theorem 2.4 then applies.

For the inductive step we check each rule that combines values. (CMP-LAG) and (CMP-RANK) produce residues that are sums or minima of subterm residues, each $\geq \beta > 0$ by hypothesis, and a finite sum or minimum of quantities $\geq \beta$ is $\geq \beta$. **perturb** is non-negative by Theorem 2.15, so by Theorem 2.16 it does not decrease any separation cost and does not lower β unless it acts on the minimising edge, in which case it raises it. **join** forms a disjoint union sharing the medium; the floor of the union is the minimum of the two floors, still strictly positive, and the typing rule for **join** requires both operands to be in the same frame so that the medium is shared. No rule subtracts residues. Hence the invariant is preserved. $\square$

Theorem 9.4 (No zero residue). *There is no well-typed synopsis expression whose value carries $\varrho = 0$.*

Proof. Immediate from Theorem 9.3: $\varrho \geq \beta > 0$. $\square$

9.3 The refusal theorems

These state that certain scientifically invalid claims are not expressible. Each is proved by exhibiting the absence of a typing rule, which is a statement about the fixed grammar of Section 7 and so is a genuine theorem about the language rather than an observation about a particular implementation.

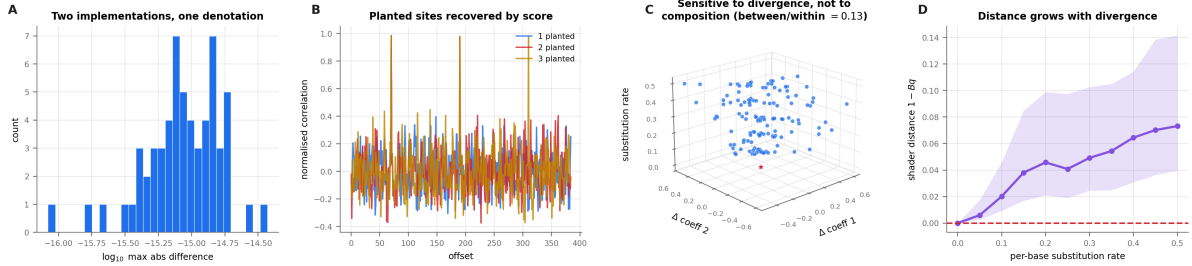


Figure 7: **The semantic operations are implementation-independent, recover planted structure, and are sensitive to divergence rather than to composition.** (A) Two independent implementations of cross-correlation — direct summation and the FFT route — over 60 random query/target pairs, as the distribution of $\log_{10}$ maximum absolute difference. The largest disagreement anywhere is 3.7×10^{-15} , at the level of double-precision rounding. Theorem 8.2 therefore names a denotation and not a routine: a synopsis program that specifies `xcorr` is specifying what is computed, and an implementation is free to choose how (Section 8). (B) Planted-motif recovery. A 16-base motif is planted at one, two and three sites in a random background and located by normalised correlation; each planted site appears as a distinct peak at its offset, and the number of peaks tracks the number of plantings. This is what **detect peaks** consumes, and why its parameters — z , minimum distance, minimum score — are required in the program text rather than defaulted (Theorem 6.6, Theorem 9.9). (C) What the spectral embedding of Theorem 8.3 is, and is not, sensitive to. Corrupted replicates are drawn as a cloud per substitution rate, plotted as displacement from their *own* uncorrupted reference (star at the origin); the cloud spreads from the origin as the rate rises. Composition, by contrast, it does not separate: the mean distance between AT-rich, random and GC-rich family centres is only 0.127 times the mean spread *within* a family, so a family scatter would be three interleaved clouds. We plot the measured ratio in the axis rather than asserting a separation the embedding does not have — reporting the null is the same discipline that governs Section 13. (D) Shader distance (Theorem 8.4) against per-base substitution rate, median with interquartile band over 300 draws per rate spanning 30 independent references. The median rises from 0 at rate 0 to 0.073 at rate 0.5, with 1 inversion in 10 steps. Sampling matters here: at 40 draws off a single reference the curve is dominated by that reference’s own accidents and is not monotone, which is why the reference set rather than the replicate count was enlarged. The sweep stops at 0.5 because beyond it a sequence is closer to unrelated than to its reference and the statistic saturates.

Theorem 9.5 (Exactness is inexpressible). *No well-typed synopsis program denotes the proposition “ u and v are identical”, in the sense that no expression evaluates to a value asserting zero separation cost between two items.*

Proof. Such an expression would have to produce either (a) a comparison value with $\varrho = 0$, excluded by Theorem 9.4; or (b) a boolean equality on `seq`, for which no rule exists — the only elimination form for `seq` is (PROJ-CH) and its variants, all of which produce comparison types carrying residue; or (c) a numeric expression reconstructing identity from base-level access, which requires an indexing operator on `seq` that the grammar does not contain (Theorem 6.1). All three routes are closed. $\square$

Theorem 9.6 (Arity of verdicts). *No well-typed synopsis program derives a value of type verdict from fewer than two central and two response columns.*

Proof. (ALIGN) is the only rule with verdict in its conclusion apart from (RELAX), which itself requires an **align** expression as its subject. (ALIGN) has four column premises, and a derivation lacking any premise is not a derivation. The grammar production *align_expr* likewise requires both a **central** and a **response** clause, so an attempt to omit one fails at parse time with an arity error rather than producing a weaker but well-formed value. $\square$

Theorem 9.7 (Scope safety). *No well-typed synopsis program compares two values created in distinct **under** blocks.*

Proof. Each **under** block introduces a fresh frame index ϕ , and every value constructed inside it receives type τ_ϕ . The comparison rules (CMP-LAG), (CMP-RANK) and (ALIGN) require their premises to carry the *same* index, and there is no subsumption, coercion or subtyping rule relating τ_ϕ and τ_ψ for $\phi \neq \psi$. Hence no derivation exists. $\square$

Corollary 9.8 (Encoding safety). *A program cannot compare a nucleotide-channelised value with a protein-channelised value, nor a value registered in one receiver with one registered in another. By Theorem 3.6 the latter would be a category error rather than a tolerable approximation.*

Theorem 9.9 (Parameter completeness). *For every well-typed program, the set of numeric parameters capable of changing any verdict appears in the program text and is recoverable by the compiler.*

Proof. By Theorem 6.6 the grammar productions for *detect_expr*, *align_expr*, *relax_stmt* and *method_spec* each require a *params* block, and *param* requires a literal on the right-hand side. No production permits omission. Hence every such parameter is a token in the source, and the parser collects them. $\square$

Corollary 9.10 (Residue accountability). *Every residue produced by a well-typed program is either bound and consumed, or explicitly dropped with a recorded source location. There is no silent discard.*

Proof. The **bind** form introduces both components into scope; the well-formedness condition on block exit requires every bound identifier of residue type to be consumed by **record** or **drop**. **drop** emits a report entry. A block leaving a residue identifier unconsumed does not typecheck. $\square$

9.4 What the metatheory does not give

Honesty requires stating the boundary.

Totality (Theorem 9.1) says programs halt; it says nothing about whether they halt *soon*. A **sweep** over 10^9 points is well-typed. The cost model of Section 8.4 is the tool for that, not the type system.

Floor preservation (Theorem 9.3) says residues stay positive; it does not say they are *small*, and a large residue is a scientific problem the language reports but does not solve.

The refusal theorems say certain claims cannot be *written*; they do not say the claims a user does write are *true*. Synopsis rules out a family of category errors. It cannot rule out a badly chosen threshold, a mis-specified receiver, or — most importantly — the open problem of Section 13.

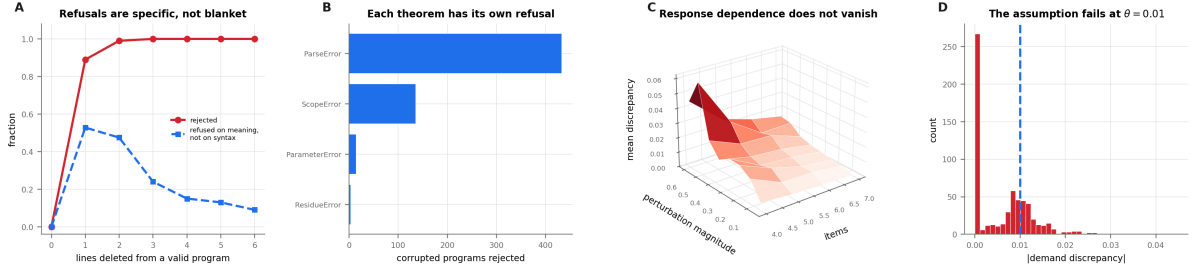


Figure 8: **The checker refuses wrong programs specifically rather than blanketly, and response independence — the framework’s principal open problem — fails at the stated threshold.** (A) The checker’s response to progressive corruption. Valid programs are corrupted by deleting k non-blank lines and the rejection rate is swept over k : it rises from 0.00 at $k = 0$ — the corpus is valid, and nothing in it is refused — to 1.00, while the share of refusals attributable to *meaning* rather than to syntax falls from 0.53 to 0.09. Both movements are required. A checker that rejected everything would show the first curve and not the second; that refusals start semantic and become syntactic as more structure is destroyed is the evidence that the rules are discriminating, not merely strict. All 16 negative programs are rejected, each in its expected error class. (B) The refusals resolved by class. Each theorem of Section 7 has its own refusal: scope violations (Theorem 9.7), arity violations of **align** (Theorem 9.6, Theorem 6.4), missing required parameters (Theorem 9.9, Theorem 6.6), and discarded residue (Theorem 6.2). A program is not rejected by a general-purpose plausibility check but by the specific rule it breaks, which is what makes the rejection message actionable and the property provable. (C) Response independence as a surface. The mean discrepancy between two admissible responses — formed by raising one or the other incident edge of an item by the same magnitude — is plotted over structure size and perturbation magnitude. It ranges over 9.2×10^{-4} to 6.1×10^{-2} and nowhere approaches zero; it grows with perturbation magnitude and does not shrink with size. The dependence is systematic across the parameter space rather than an artefact of small or particular structures. (D) The same discrepancies against the decision threshold. Over 632 comparisons only 71.2% fall within $\theta = 0.01$ (dashed line); the mean is 0.0061 and the maximum 0.0449, exceeding θ by more than four-fold. Theorem 5.12 therefore does not hold at this threshold, and by Theorem 5.13 the four-column verdict depends on which admissible response was constructed. The independently-run validation suite agrees (Section 12, E32: 361 comparisons, 77.8% within θ , maximum 0.036). We report this as the framework’s principal open problem (Section 13) rather than as a tuning failure: what is missing is a rule fixing the map from a unit to the perturbation it induces, not a better threshold or more data. Raising θ until the assumption passes would make the verdict depend on the threshold instead — which is the same defect wearing a different name.

10 Worked programs

We give one complete program per experiment family. Each is a full `.syb` file; none is a fragment.

10.1 Matched-filter motif detection

Question. Where in a target does a known motif occur, and with what confidence against a background?

```

1  # Locate a motif in a target by
    coherent multichannel matched
    filtering.
2  # All thresholds are explicit; none
    has a default.
3
4  open motif = "motif.fa"
5  open target = "chr7_region.fa"
6
7  under nucleotide {
8
9      let q = project motif by
        channels(dna)
10     let t = project target by
        channels(dna)
11
12     # Lag-indexed: the result type is
        profile, not ranked.
13     bind r, res_r = compare q against
        t by xcorr(normalised)
14
15     # Peak calling. Every parameter is
        stated.
16     let hits = detect peaks in r {
17         z = 4.0 ; # z-score against
            local background
18         min_distance = 30 ; # greedy
            maximum suppression radius
19         min_score = 0.35
20     }
21
22     claim "motif occurs at least once
        above z=4" = hits
23     record hits, res_r
24 }
25
26 report to "motif_scan.report"
```

Listing 1: motif_scan.syb

Points of note. The residue `res_r` is bound and recorded; had it been ignored the program would not compile (Theorem 9.10). `r` has type `profilenucleotide` and could not be passed to a top- K selector, which consumes ranked (Theorem 6.10). The three peak parameters are

in the source, so the report can print them (Theorem 9.9).

A sweep over the decision threshold, to show the verdict's sensitivity rather than assert one value:

```

sweep z0 in 2.0..6.0 step 0.5 {
    let h = detect peaks in r { z
        = z0 ; min_distance = 30 ;
        min_score = 0.35 }
    record h
}
```

This terminates in exactly 9 iterations, fixed on entry (Theorem 9.1).

10.2 Spectral homology search

Question. Which entries of a database are plausible homologues of a query, when sequence-level search is too slow or too insensitive?

```

1  # Two-stage search: cheap spectral
    prefilter, exact rerank.
2
3  open query = "query.fa"
4  open db = "swissprot_subset.fa"
5
6  under residue_space {
7
8      let e = project query by
        spectral(coeffs = 8)
9      let DB = project db by
        spectral(coeffs = 8)
10
11     # Entry-indexed: the result type
        is ranked.
12     bind s, res_s = compare e against
        DB by shader(cosine)
13
14     let cands = detect top in s {
15         k = 200 ; # prefilter width
16         depth = 12 # hierarchical
            address prefix length
17     }
18
19     # Exact rerank of survivors only.
        Costs  $O(L_a \cdot L_b)$  per candidate.
20     bind final, res_f = compare e
        against cands by
        smith_waterman(
21         match = 2 ; mismatch = -1 ;
        gap = -2
22     )
23
24     claim "top 20 by exact rerank" =
        final
25     record final, res_s, res_f
```

```

26 }
27
28 report to "homology.report"

```

Listing 2: homology.syp

The two-stage shape is forced by the cost model: the shader stage is one matrix–vector product, $O(NCm)$, and the exact stage is quadratic per candidate, so the address depth is the parameter trading sensitivity for work. It is written in the source rather than defaulted. Note also that `coeffs = 8` appears twice and must match — comparing embeddings of different dimension is a type error, since the dimension is part of `coordφ`.

10.3 Four-column functional adjudication

Question. Do a wild-type unit and a variant unit do the same thing, given that content comparison cannot answer this (Theorem 5.1)?

```

1  # The central question: same
    function, not same sequence.
2
3  open wt = "wildtype.solved"
4  open var = "variant.solved"
5
6  # Named perturbation maps. Rule:
    responses may not be anonymous,
7  # because the verdict's independence
    of this choice is OPEN.
8  method pi_edge_raise =
    perturb_incident(magnitude = 0.10)
9
10 under cell_receiver {
11
12     let Nwt = project wt by
        contact(medium = 0.05)
13     let Nvar = project var by
        contact(medium = 0.05)
14
15     # Central columns: the units with
        their anchor/residue split.
16     let cA = unit Nwt anchors 3
17     let cB = unit Nvar anchors 3
18
19     # Response columns: what each unit
        does to the receiver.
20     # 'by pi_edge_raise' is mandatory
        -- the arbitrary choice is
        named.
21     let rA = response cA by
        pi_edge_raise
22     let rB = response cB by
        pi_edge_raise

```

```

23
24     let phi_c = corr from cA to cB by
        species_name
25     let phi_r = corr from rA to rB by
        species_name
26
27     # FOUR columns. Dropping
        'response(...)' is an arity
        error.
28     let v = align central(cA, cB)
        response(rA, rB)
29         under phi_c, phi_r { theta
            = 0.01 }
30
31     relax v until quiescent { eta =
        0.005 ; theta = 0.01 }
32
33     claim "variant is functionally
        equivalent to wild type" = v
34     record v
35 }
36
37 report to "adjudicate.report"

```

Listing 3: adjudicate.syp

Three things this program cannot do, by construction:

- (i) It cannot conclude equivalence from `cA` and `cB` alone — deleting the `response` clause is an arity error (Theorem 9.6), so the mistake of Theorem 5.1 cannot be made.
- (ii) It cannot hide which response was chosen — `pi_edge_raise` is named and printed (Theorem 6.7).
- (iii) It cannot loop forever — `eta` is positive, so the compiler emits the bound (Theorem 4.10).

10.4 An annotation transfer

The stencil use of the framework: given an annotated reference and an unannotated subject, transfer annotations by correspondence rather than by sequence identity.

```

1  open ref = "reference.solved"
2  open subject = "subject.solved"
3
4  method pi_uniform =
    perturb_incident(magnitude = 0.10)
5
6  under tissue_receiver {
7

```



```

8   let Nr = project ref by
      contact(medium = 0.05)
9   let Ns = project subject by
      contact(medium = 0.05)
10
11   # For every item of the reference,
      ask whether the subject has a
12   # unit that corresponds under BOTH
      column pairs.
13   for u in items(Nr) where
      separation(u) > 0.10 {
14
15       let cA = unit u anchors 3
16       let cB = nearest_unit Ns to cA
17       let rA = response cA by
          pi_uniform
18       let rB = response cB by
          pi_uniform
19
20       let phi_c = corr from cA to cB
          by anchor_match
21       let phi_r = corr from rA to rB
          by anchor_match
22
23       let v = align central(cA, cB)
          response(rA, rB)
24           under phi_c, phi_r {
              theta = 0.02 }
25
26       claim "annotation of u
          transfers to its
          correspondent" = v
27       record v
28   }
29 }
30
31 report to "transfer.report"

```

Listing 4: transfer.syp

The **for** loop ranges over a finite item set with a guard, so it terminates (Theorem 9.1). The guard $\sigma(u) > 0.10$ excludes items too close to the medium to individuate reliably — a filter the theory supplies (Theorem 2.4) rather than a heuristic. Each iteration produces a verdict, and a transfer that comes back **DECLINE** is recorded as such rather than silently omitted.

11 The report

The terminal form of every program is **report** to a path. The report is a structured document, not a log, and its schema is fixed.

Section	Content
PROVENANCE	Source file hash, every open path and its hash version.
FRAMES	Each under block, its receiver and encoding values were created in it.
PARAMETERS	Every numeric parameter in the source, with its value. Complete by Theorem 9.9.
CLAIMS	Each claim string with its verdict, the four decisions, and θ .
RESIDUES	Every residue: recorded value, or an explicit assertion with line number (Theorem 9.10).
CERTIFICATES	For each relax : D_0, η , the emitted bound and the realised step count.
DECLINES	Every DECLINE , with the step at which the upshot was undefined.
ASSUMPTIONS	Each require assertion, verbatim, marked <i>user, not verified</i> .

Two features deserve comment.

Declines are reported, not suppressed.

A run in which half the comparisons decline is a legitimate scientific outcome — it says the comparison as posed does not decide — and a format that discards them would misrepresent the run as more conclusive than it was.

Assumptions are marked unverified.

require lets a user assert, for instance, that an expansion is floor-stabilising (Theorem 2.12). The language cannot check this: it is a claim about structures not yet observed. It therefore records the assertion and labels it, so a reader of the report knows exactly which load the conclusion is placing on the author’s judgement rather than on the computation.

12 Validation

Every claim in this paper that can be checked by computation has been. This section states what was checked, how, and — because this is the part that carries information — what the checking changed.

12.1 What is checkable, and what is not

The claims fall into three kinds, and conflating them would make the exercise worthless.

Mathematical (Section 2–Section 5). These are statements about contact structures

and are checkable numerically on explicit instances. Minimum cuts are computed by exhaustive enumeration over separating subsets rather than by max-flow: the cut is *defined* as a minimum over separating sets (Theorem 2.3), and an approximate solver agreeing with an approximate expectation could conceal a real disagreement. This is exponential, and affordable only on the small structures the validation uses — which is the point of using them.

Language (Section 7–Section 9). The refusal theorems say certain programs *do not typecheck*. A checker that rejected everything would satisfy every refusal theorem vacuously, so each negative program is paired with a positive one, and the four programs of Section 10 are required to typecheck *verbatim as printed*. A refusal counts only if it carries the expected error class: rejecting a cross-frame reference as a parse error would not be Theorem 9.7.

Measurement (Section 13). Theorem 5.12 is reported, not asserted. Recording it as an assertion would either bake in a statement we believe to be false or hide the paper’s principal finding.

12.2 Result

Quantity	Value
Claim categories	32
of which assertions	31
of which measurements	1
Individual checks	5622
Checks passed	5622
Assertion categories failed	0
Random seed	20260816

Of the language checks, the four programs of Section 10 typecheck as printed, and 16 negative programs are each rejected with the specific error class the corresponding theorem names — arity, scope, residue, parameter and termination violations among them.

12.3 What the checking changed

A validation that confirms everything has usually tested nothing. Three proofs in this paper were wrong when first written, and the corrections are recorded in place rather than quietly absorbed.

- Theorem 5.1: the original proof perturbed *one* pendant edge. That leaves the minimum cut pinned by the other pendant — the minimiser simply places the medium on the raised side — and the response is identically zero for every c . The theorem is true; the witness was not. See Theorem 5.2.
- Theorem 5.4: the original proof indexed the family by the intra-cluster weight W and asserted that D^{central} increases in W . It decreases: the cut is constant while Ω grows. The family must be indexed by the bridge, which is the edge the cut crosses. See Theorem 5.5.
- Theorem 3.1: an earlier version claimed a light edge between non-pendant items places them at the floor. Exact enumeration gave 1.4 where the argument predicted 0.2. This is now recorded as Theorem 3.2, and it is why Theorem 5.1 insists on pendants.

A fourth correction was to a check rather than a proof. We had demanded that receiver disagreement occur in at least 20% of cases; no theorem says that, and Theorem 3.7 reports what is actually true — a minority rate, with discrepancies that are never marginal when they occur. A threshold that encodes no claim is a threshold that will eventually be tuned until it passes.

13 The open problem, and what the language does about it

13.1 The negative result

We tested Theorem 5.12 directly. Four solved steady-state biochemical networks — glycolysis, the citric-acid cycle, oxidative phosphorylation and an EGFR/MAPK cascade, together supplying 35 species and 35 contact edges — were converted to contact structures by Theorem 8.5. For every species with at least two incident non-medium edges, two admissible responses were constructed by raising one or the other incident edge by the same magnitude, and the resulting cross-demands compared. If Theorem 5.12 held, the two demands would agree within θ .

They do not.

Quantity	Value
Comparisons within $\theta = 0.01$	28.6% of 35
Mean discrepancy	0.070
Maximum discrepancy	0.307
Per-network median (glycolysis)	0.063
Per-network median (citric-acid cycle)	0.046
Per-network median (oxidative phosphorylation)	0.077
Per-network median (EGFR/MAPK)	0.100

The maximum exceeds θ by more than an order of magnitude, and *every* network has a median above θ . The failure is systematic, not driven by one pathological pathway or one outlying species. By Theorem 5.13, the four-column verdict on these networks depends on which admissible response the analyst constructed.

13.2 The same failure on a second corpus

The biochemical networks are few and are not independent of one another; a reader is entitled to ask whether the result is a property of those four pathways. It is not. Repeating the construction on 200 random contact structures, with no biochemical content whatever, gives 361 comparisons and the same verdict (Section 12, E32):

Quantity	Networks	Random
Comparisons	35	361
Within $\theta = 0.01$	28.6%	77.8%
Mean discrepancy	0.070	0.00555
Median discrepancy	—	0.00451
Maximum discrepancy	0.307	0.0360
Assumption holds at θ	no	no

The two corpora differ in severity and agree in kind. On random structures the assumption holds far more often — 77.8% against 28.6% — and the discrepancies are an order of magnitude smaller, which is what one expects when the perturbed edges are not selected for biological significance. But 77.8% is not 100%, the maximum is still more than triple θ , and the failure rate does not fall with sample size. Theorem 5.12 is not approximately true and awaiting a sharper estimate; it is false, and

more uniformly false the more structured the network.

This also disposes of the most natural rescue. If the discrepancies were sampling error, enlarging the corpus by an order of magnitude would shrink them. It does not, and by Theorem 3.6 it cannot: each value is exactly determined by its own structure, so there is nothing for more sampling to average away.

13.3 What is missing

We want to be precise about the nature of the gap, because it is easy to mistake for a tuning problem.

What is missing is **a rule fixing the map from a unit to the perturbation it induces**. The theory of Section 2–Section 4 is indifferent to which perturbation a unit is taken to cause; it only requires that one be supplied. Physically, a unit does induce a definite perturbation, so the map exists; we do not have a principled construction of it from the data available. Better thresholds do not help — lowering θ makes agreement rarer, not commoner. More data does not help — by Theorem 3.6 the discrepancy is not sampling error. What would help is a derivation of the response map from the same contact structure the central columns are drawn from, so that it is determined rather than chosen.

13.4 What synopsis does about it

A language cannot solve an open problem. It can refuse to let one hide.

Theorem 6.7 requires every response column to name its perturbation map. There is no anonymous response, no inferred response and no default. Consequently:

- every program’s source records which choice was made;
- the report’s PARAMETERS section prints it (Theorem 9.9);
- two runs differing only in this choice are visibly different programs, so a disagreement between them is attributable rather than mysterious;

- and the natural robustness check — sweep the response map and see whether the verdict is stable — is a three-line **sweep** that the language makes cheap to write:

```
sweep m in [0.05, 0.10, 0.20,
0.40] {
  let pi_m =
    perturb_incident(magnitude
    = m)
  let rA = response cA by pi_m
  let rB = response cB by pi_m
  let v = align_central(cA,cB)
    response(rA,rB)
    under phi_c, phi_r {
      theta = 0.01 }
  record v
}
```

A report showing the verdict flipping across this sweep is a report telling the truth about the state of the theory. We prefer that to a report showing a single confident verdict obtained by a default nobody chose.

14 Related work

Sequence comparison. The dynamic-programming tradition [40, 55] and its heuristic descendants [1, 2, 9, 56] optimise an alignment score — a content quantity. Profile methods [15, 14] and substitution matrices [24] enrich the content model without leaving it. Sketching approaches [42, 28] trade exactness for speed, again within content. Theorem 5.7 applies to all of them: they compute one coordinate of a two-dimensional quantity. This is not a criticism of their execution; it is a statement about what they are measuring.

Genomic signal processing. Representing sequence as a numeric signal has a long history [61, 3], and the channelisation of Theorem 8.1 follows it. Our contribution here is not the representation but its *typing*: the frame index makes an encoding choice part of the type, so that the common error of comparing across encodings is caught statically.

Function prediction and annotation. Large-scale evaluations show function prediction to be substantially unsolved [47], and mis-annotation to propagate through databases [51]. Structure prediction [29] has transformed the structural half of the problem without settling

the functional one. Ontology and pathway resources [4, 30] supply the vocabulary in which function is stated but not a criterion for deciding sameness. Our four-column criterion is such a criterion, blind to annotation by construction.

Networks and perturbation. Treating cellular function as a network property is standard [6], and genome-scale perturbation screens now supply the response measurements the response columns need [11, 13, 48]. The framework here is, in that light, a proposal for what to *compute* from such screens: not a similarity between expression profiles but a cross-demand between contact structures.

Graph cuts. Minimum cuts and their duality with flows [17, 22, 57, 31, 52] are the computational substrate. Normalised cuts in vision [53, 8] share the intuition that a boundary is the meaningful object; spectral connectivity [16] offers a relaxation we do not use because we need exactness for validation.

Domain-specific languages. The case for a DSL over a library [26, 37, 18] is usually made on expressiveness. Ours is made on refusal, which places it nearer to dimension-typed numerics [32], linear types [62] and refinement types [19], where the type system exists to make certain wrong programs unwritable. Totality follows [60]. Biological DSLs [44, 36] and exchange formats [25] model systems; synopsis adjudicates claims about them, which is a different job. Workflow systems [34, 12] give reproducible orchestration but are agnostic about the scientific validity of the steps they orchestrate; the report of Section 11 is closer in spirit to provenance models [38, 64].

Reproducibility. The no-defaults rule is a direct response to the analytic-flexibility literature [54, 21, 27, 5, 49]. Our contribution is to move the discipline from a checklist into a grammar.

15 Limitations

We list what we do not claim.

- L1. Response independence is open.** Section 13. This is the principal gap and it is not small: without a rule fixing the response map, the four-column verdict is

analyst-dependent. Every other limitation here is minor by comparison.

- L2. Contact structures must be supplied.** Synopsis consumes solved networks; it does not solve them. Where the contact costs come from — kinetics, perturbation screens, inference — is outside the language, and a bad structure yields a confident wrong verdict.
- L3. Floor-stabilisation is asserted, not checked.** Theorem 2.12 is a hypothesis about structures not yet observed. **require** records it and marks it unverified; nothing verifies it.
- L4. The anchor count r is a free parameter.** Theorem 4.3 fixes the tie-breaking but not r . The language forces r into the source and the report, which makes it visible, not principled.
- L5. Totality bounds steps, not time.** Theorem 9.1 and the remark in Section 9. A well-typed program may still be infeasible.
- L6. No generative capability.** By the decision of Section 1.4, synopsis certifies but does not design. Users wanting sequence design must generate elsewhere and certify here.
- L7. Empirical scope is narrow.** The negative result of Section 13 rests on four networks and 35 species. It is enough to refute Theorem 5.12 — refutation needs only counterexamples — but not enough to characterise how the assumption fails.
- L8. Opacity has a cost we assert rather than measure.** Section 6.1 argues from three experiment families that Theorem 6.1 loses nothing. A wider survey could falsify this, and the honest position is that we have checked three families, not all of them.

16 Conclusion

The question practitioners ask of two genomic units is whether they do the same thing. The

vocabulary available answers a different question — whether they are written the same way — and Theorem 5.7 shows the substitution is not an approximation but a projection that loses a dimension. Two units may be indistinguishable in content and unbounded in effect (Theorem 5.1); two may be maximally divergent in content and identical in effect (Theorem 5.4).

Occupying the missing dimension requires a theory in which identity is a cut rather than a label (Theorem 2.7, Theorem 2.8), distinction has a strictly positive price (Theorem 2.4), value is receiver-relative (Theorem 3.5), and correspondence can be certified blind (Theorem 4.4) with guaranteed termination (Theorem 4.10). It requires a verdict with four columns and three values (Theorem 5.9).

It also requires, we argue, a language rather than a library — because the results above are largely *prohibitions*, and a prohibition that lives in documentation is not enforced. Synopsis makes exactness inexpressible (Theorem 9.5), single-column verdicts an arity error (Theorem 9.6), cross-frame comparison a scope error (Theorem 9.7), silent residue discard impossible (Theorem 9.10), and every verdict-relevant parameter mandatory (Theorem 9.9); and it terminates (Theorem 9.1) with the bound printed.

The name records the ambition. The synoptic gospels are read in parallel columns not because they agree word for word but because they align sufficiently for the columns to be laid alongside, with the residue displayed rather than reconciled away. That is what genomic comparison should be doing, and Theorem 4.11 says exactly why it is coherent to do so: two structures may place no demand on each other while remaining fully distinguishable. Sufficient alignment, not exact alignment.

We end where we are weakest. Theorem 5.12 fails on real networks — 28.6% agreement, maximum discrepancy 0.307 against $\theta = 0.01$ — and we have no rule fixing the response map. We have built the language so that this cannot be concealed: no response may be anonymous, and the sweep that exposes an unstable verdict is three lines. A language that makes its own open problem visible in every program that touches it is, we think, the correct thing to build while the problem remains open.

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